# A NEXT GENERATION AI FRAMEWORK FOR INTELLIGENT TRAVEL SUPPORT IN SRI LANKA

## COMMON INTEGRATED SOLUTION REPORT

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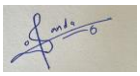

October 2026

# DECLARATION AND SUPERVISOR CERTIFICATION

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# ABSTRACT

Tourism is vital to Sri Lanka's economy, yet the information that would help visitors, drivers and destination managers make good decisions is scattered across separate tools and arrives too late. Itineraries are static, managers cannot observe how visitors feel during a visit, transport safety supervision is reactive, and crowding at popular sites is managed by intuition. This report presents an integrated, AI-based framework for intelligent travel support in Sri Lanka, delivered as a mobile application and web dashboards and developed as four complementary components with a shared integration layer. Component C1, an AI conversational travel analyzer, extracts multiple intents and constraints from text and voice requests, generates personalized packages using route-ranking and rule-based selection,and supports itinerary adaption in response to changing travel conditions.Component C2 uses a custom CNN trained on RAF-DB to classify five facial emotion classes. It provides zone-wise satisfaction analysis, tracks changes in emotion over time, and recommends places based on users' selected interests. Component C3 fuses driver fatigue monitoring and forward collision risk into a Composite Safety Index on an ESP32 edge hub, issuing tiered alerts while keeping video on the vehicle. Component C4 forecasts crowding across attractions and recommends redistribution of visitors. A shared spatial model, versioned data contracts, an event bus and consistent consent and pseudonymous identity connect the four components. The work follows a design science approach and is evaluated through component benchmarking, ablation, scenario testing, three cross-component scenarios, device and load tests, usability assessment and a pilot at a Sri Lankan heritage site. The study contributes a scalable, privacy-preserving integrated architecture and a closed-loop design in which the output of one component informs the decisions of another, in support of SDG 3, SDG 8 and SDG 11.

**Keywords:** smart tourism, emotion recognition, conversational AI, multi-objective optimization, driver safety

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# LIST OF ABBREVIATIONS

| Abbreviation | Full form |
|---|---|
| AI | Artificial Intelligence |
| API | Application Programming Interface |
| BERT | Bidirectional Encoder Representations from Transformers |
| CLAHE | Contrast Limited Adaptive Histogram Equalization |
| CNN | Convolutional Neural Network |
| DSR | Design Science Research |
| EAR | Eye Aspect Ratio |
| FER | Facial Emotion Recognition |
| GDP | Gross Domestic Product |
| GIS | Geographic Information System |
| GPS | Global Positioning System |
| GRU | Gated Recurrent Unit |
| INT8 | 8-bit Integer (quantization format) |
| LSTM | Long Short-Term Memory |
| MAE | Mean Absolute Error |
| MAPE | Mean Absolute Percentage Error |
| MLP | Multi-Layer Perceptron |
| NLP | Natural Language Processing |
| NSGA-II | Non-dominated Sorting Genetic Algorithm II |
| PDPA | Personal Data Protection Act |
| RAF-DB | Real-world Affective Faces Database |
| REST | Representational State Transfer |
| SDG | Sustainable Development Goal |
| SLIIT | Sri Lanka Institute of Information Technology |
| SUS | System Usability Scale |
| TFLite | TensorFlow Lite |
| TLS | Transport Layer Security |
| UI | User Interface |
| UNESCO | United Nations Educational, Scientific and Cultural Organization |
| UX | User Experience |
| YOLO | You Only Look Once (object detector) |

# 1 INTRODUCTION AND PROJECT CONTEXT

## 1.1 Background and context

### 1.1.1 Tourism in Sri Lanka and the need for intelligent travel support

Tourism is one of the most important sectors of the Sri Lankan economy. In the years before the 2020 pandemic the country received roughly two million international visitors annually, with arrivals peaking at about 2.3 million in 2018, and the industry contributed an estimated 4 to 5 percent of gross domestic product while supporting several hundred thousand direct and indirect jobs [1]. Cultural attractions are central to this performance. The island hosts eight UNESCO World Heritage Sites, including Sigiriya, the ancient city of Polonnaruwa, the sacred city of Kandy and the Old Town of Galle and its fortifications, together with national parks, beaches, tea-country railways and many nationally significant temples and archaeological reserves. After the disruption of the pandemic and the economic crisis that followed, the recovery and the long-term sustainability of the sector depend on delivering higher-quality, better-managed visitor experiences rather than on volume alone.

The journey of a visitor to Sri Lanka is, however, fragmented. Before arrival the visitor searches websites, forums and social media to decide where to go and how long to stay. During the trip the visitor moves between destinations by road, often with a hired driver, on a network that has heavy traffic, winding hill roads and a high rate of road traffic injuries. At each destination the visitor meets crowds, heat, queues, uneven signage and limited interpretation. At every stage the information that would help the visitor, the driver, the destination manager and the tourism authority to make a better decision exists somewhere, but it is not captured, joined together or acted upon while the trip is still in progress.

### 1.1.2 The four stages of the tourist journey

The project described in this report starts from the observation that a trip can be divided into four stages, each of which suffers from a different kind of missing intelligence:

- **Planning.** Visitors build itineraries from generic packages or from many separate websites. Existing tools do not understand a request such as "a relaxed five-day trip for a family with an elderly parent, with a budget of 1,200 dollars, a little culture and no long drives" and rarely trade off competing objectives such as cost, time, interest match and comfort. Itineraries are also static: they do not change when the weather, the traffic or the mood of the group changes.
- **Experience at the destination.** Managers of heritage sites cannot observe how visitors feel while a visit is in progress. A congested stairway or a poorly designed exhibit may generate dissatisfaction for weeks before it appears in an online review.
- **Transport between destinations.** Tourists spend a large share of their holiday in vehicles. Driver fatigue, distraction, harsh driving and risky road segments directly affect safety, yet safety supervision is mostly reactive and relies on after-the-fact incident reports.
- **Flow across the destination network.** A small number of famous sites absorb most of the visitors at peak hours, while equally rewarding sites nearby stay quiet. Overcrowding degrades the experience, damages fragile heritage and raises safety risk, but managers lack short-term forecasts and redistribution tools.

### 1.1.3 Smartphones, on-device AI and smart tourism

The concept of smart tourism describes the use of information and communication technology to create, manage and deliver intelligent tourism experiences through the collection and exploitation of data [2], [3]. Smartphones are the natural platform for such systems. Almost every international tourist carries a device with a camera, a GPS receiver, motion sensors, a microphone and enough processing power to run compact neural networks locally. Advances in lightweight convolutional architectures [4], quantized inference and large language models make it feasible to build services that understand natural-language travel requests, interpret the visitor's state and give timely

guidance, without requiring expensive proprietary hardware at the destination. For vehicles, it utilizes a low-cost, custom IoT edge unit that keeps raw video strictly on-device, processing sensitive signals locally before sending status data through Firebase to the driver's smartphone app.

### 1.1.4 The overall project

This report presents the group project "A Next-Generation AI Framework for Intelligent Travel Support in Sri Lanka" (project code R26-IT-112). The framework consists of four independent but complementary components, each of which is the subject of an individual dissertation, and a shared integration layer that is the subject of this common report.

**Table 1.1.1: The four components of the framework and the team members responsible**

| Component | Title | Student (ID) | Tourist-journey stage |
|---|---|---|---|
| C1 | AI conversational travel analyzer with intelligent package generation and a real-time adaptive engine | Senevirathna T.M.N.D.L(IT22082442) | Planning and trip adaptation |
| C2 | Adaptive emotion recognition for tourist experience management | Kaweeshwara M.D.O (IT22216014) | Experience at the destination |
| C3 | Smart driver and vehicle proactive safety system | Rupasinghe M.L.K.S. (IT22176974) | Transport between destinations |
| C4 | Intelligent tourism flow optimization system | Manthrirathne M.A.A.C (IT22607614) | Flow across the destination network |

## 1.2 Stakeholders and user needs

The framework serves six groups of stakeholders: tourists, drivers, destination managers, site staff, transport operators and tourism authorities. Each group has different interests and different influence over adoption. The analysis below records

how the design responds to each group, and the needs it implies are carried into the functional and non-functional requirements in Chapter 4.

**Table 1.2.1: Stakeholder analysis**

| Stakeholder | Interest in the system | Influence | Main needs | How the design responds |
|---|---|---|---|---|
| Tourists | Enjoyable, safe and well-planned trip | High (adoption) | Ease of use; privacy; useful suggestions | Conversational planner; granular consent; on-device processing; concise suggestions |
| Drivers | Safe working conditions; fair treatment | High (acceptance) | Non-intrusive alerts; no punitive use of data | In-cabin buzzer and voice alerts; limited reporting; welfare framing |
| Destination managers | Satisfaction, safety and protection of the site | High | Timely, actionable and explainable insight | Heatmaps, evidence, recommendations, configurable thresholds |
| Site staff | Manageable workload and safety | Medium | Relevant alerts without overload | Rate-limited and de-duplicated alerts with evidence |
| Transport operators | Safe fleet; reputation; cost | Medium | Visibility of trip safety and vehicle condition | Trip summaries; urgent alerts; vehicle indicators |
| Tourism authorities | Sector growth and sustainability | High (policy) | Aggregated, reliable statistics | Anonymized exports with sample counts |
| Local communities and religious custodians | Respect for sacred places and local life | Medium | No intrusion at sacred areas | No-capture zones; photography reminders |

| Stakeholder | Interest in the system | Influence | Main needs | How the design responds |
|---|---|---|---|---|
| Regulators | Compliance with data protection law | High | Lawful basis, consent and security | Consent per service; minimization; retention limits [5] |
| Research community | Reproducible findings | Low | Methods, data and code | Documented protocol; anonymized dataset; versioned models |

### 1.2.1 Significance for tourists

For the tourist, the framework promises a smoother trip. The planner reduces the effort of building an itinerary and explains the trade-offs between options. The adaptive engine means that a plan that has been made does not become a burden when the weather turns, a site closes or a queue forms. The emotion component produces a personal journey summary that the tourist can keep, and gives suggestions at the moment a visit is deteriorating. The safety component protects the traveler in the vehicle without any action on their part. Because the tourist does not have to carry additional hardware, the benefits are available to every visitor with a smartphone.

### 1.2.2 Significance for drivers and transport operators

Professional drivers in the tourism sector work long hours and are responsible for the safety of their passengers. A system that detects fatigue and recommends a break before a risky segment supports the driver's own interest in a safe trip. Operators gain visibility of trip safety and vehicle condition and can plan maintenance and rest schedules more intelligently. By framing the monitoring as support and by limiting what is reported to the operator, the design aims to gain the acceptance of drivers, which is a precondition for any benefit.

### 1.2.3 Significance for destination managers and authorities

Destination managers currently depend on lagging indicators. The framework gives them near real-time views of how visitors feel, which zones are close to capacity and

what is likely to happen in the next hours, together with recommendations and the evidence for them. This supports operational decisions such as opening an additional access route or deploying staff, and strategic decisions such as investment in shade, seating or interpretation. For tourism authorities, aggregated and anonymized analytics across sites provide a more objective basis for policy, promotion and the management of carrying capacity at heritage sites that are both economically important and physically fragile.

### 1.3 Alignment with the Sustainable Development Goals

The framework contributes to three United Nations Sustainable Development Goals. For **SDG 8 (Decent Work and Economic Growth)**, better planning, better experiences and better distribution of visitors support repeat visits, longer stays and positive recommendations, which sustain tourism revenue and local employment. For **SDG 11 (Sustainable Cities and Communities)**, evidence-based management of cultural heritage, crowd flows and public spaces helps managers to protect heritage and to distribute visitors and resources more responsibly. For **SDG 3 (Good Health and Well-being)**, and in particular the target of reducing deaths and injuries from road traffic accidents, proactive driver and vehicle safety monitoring supports safer transport for tourists and drivers. These contributions are plausible and mediated by management and user decisions. The project does not claim to measure economic, social or public-health impact directly.

### 1.4 Purpose and scope of the common report

This common report presents the integrated problem and solution of project R26-IT-112. It explains how the four components fit together through a shared spatial model, versioned data contracts, an event bus and consent services, and it reports the group-level evaluation. The detailed component research, design and experiments are reported in the four individual dissertations, which are the authoritative source for component internals. Chapter 8 summarises the conclusions, contributions and future work.

# 2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

## 2.1 Overall problem definition

The common problem is the fragmentation of intelligence across the tourist journey. Information that would help visitors, drivers and destination managers is captured in separate tools, is not joined together, and is not acted upon while a trip is still in progress. The problem is stated below separately from any proposed technology.

The core research problem addressed by the group project is stated as follows.

> *How can an integrated, mobile-first artificial intelligence framework be designed and implemented that supports tourists in Sri Lanka across the stages of planning, destination experience, road transport and crowd flow, such that the components share information through a common architecture, adapt to live conditions, protect user privacy and produce actionable outputs for tourists, drivers and destination managers?*

The problem is decomposed into one sub-problem per component and one for integration.

1. **Planning problem (C1).** How can free-text and spoken travel requests with several intents be converted into optimized, personalized packages, and how can these packages be adapted in real time as conditions change?
2. **Experience problem (C2).** How can emotion recognition run on a standard smartphone, be linked to zones of a heritage site, be modelled over a visit, be monitored for anomalies and be converted into recommendations for tourists and managers?
3. **Safety problem (C3).** How can driver fatigue and forward-collision risk be combined into a single risk index, and into timely, proportionate alerts, using a low-cost, privacy-preserving in-vehicle edge unit?
4. **Flow problem (C4).** How can crowding across a network of attractions be forecast and redistributed through recommendations that reach tourists and operators in time to matter?

5. **Integration problem.** How can the four components be connected through a common data contract, identity, consent and spatial model, so that each component can use the others' outputs while remaining independently deployable and privacy-preserving?

## 2.2 Root causes and overall research gap

This section reviews the research that underpins the four components. For each component the review covers the main approaches, the closest existing systems and the limitations that remain. The section ends with a review of integrated, multi-component tourism platforms, because the distinguishing aim of this project is that the components are designed to exchange information and to act on each other's outputs. A summary of the reviewed studies and the gaps they leave open is given in Table 2.

### 2.2.1 Conversational travel planning and package optimization (C1)

Travel planning has been studied as an itinerary recommendation and orienteering problem, in which a set of points of interest must be selected and ordered to maximize the traveler's utility subject to time and budget constraints. Classical recommender systems rely on explicit ratings or implicit click histories and perform poorly for new visitors with no history, which is the typical situation for a first-time tourist. Conversational interfaces overcome the cold-start problem by eliciting preferences through dialogue. Pre-trained transformer language models such as BERT [6] have made it practical to classify intents and extract entities such as dates, budgets, group composition and interests from free text, including multiple intents expressed in a single message.

Package generation has a multi-objective character. A tourist wants to minimize cost and travel time while maximizing interest match, comfort and diversity, and these objectives conflict. Multi-objective evolutionary algorithms such as NSGA-II [7] produce a set of non-dominated alternatives, the Pareto front, from which a final package can be chosen according to the user's stated priorities. The literature on conversational planners, however, usually treats the itinerary as fixed once it has been generated. Few systems adapt the plan while the trip is in progress in response to weather, closures, traffic, crowding or the emotional state of the travelers, and few

combine voice interaction with proactive suggestions. This limitation motivates the real-time adaptive engine in C1.

### 2.2.2 Emotion recognition for tourist experience management (C2)

The tourism and hospitality literature consistently treats emotion as a primary determinant of satisfaction, loyalty and word-of-mouth behaviors [8]. Gonzalez-Rodriguez et al. [9] connect automatic facial-expression analysis to tourist satisfaction and argue that it is a more objective measure than self-reported questionnaires, but their study used still images in a controlled setting. Kement et al. [10] show that research on facial emotion recognition in hospitality and tourism is dominated by conceptual frameworks and laboratory experiments, with very few field deployments and almost no systems that produce actionable recommendations. Kwon and Yu [11] demonstrate that real-time multi-CNN emotion recognition is feasible in a museum, using fixed institutional cameras indoors, while Qi and Han [12] detect high-arousal negative emotion in tourism emergencies, but reactively, after an emergency has developed.

Wearable and multi-modal approaches show that combining signals outperforms a single modality in naturalistic conditions. Matsuda et al [13] estimate tourist emotion and satisfaction from behavioral and audiovisual data captured by wearable devices, and Santamaria-Granados et al. [14] use physiological signals to personalize recommendations. Both depend on specialized hardware that is impractical for sites with thousands of daily visitors. Systematic reviews [15], [16], [17] report that laboratory accuracy overstates field accuracy, that late fusion is a practical option for deployment, that environmental context such as location and crowding should be included, and that applications to tourism management are almost unexplored. Barrett et al. [18] add that facial configurations do not map reliably to internal emotional states, which argues for treating a facial classification as one noisy signal to be interpreted in context. Component C2 is built on these observations: a Custom CNN model classifier [4] runs on the phone, zone-wise satisfaction mapping summarizes visitor satisfaction, and emotion trajectories show changes in detected emotions over time. The system recommends places based on users' selected interests.

### 2.2.3 Driver and vehicle safety monitoring (C3)

Road safety is a major concern for tourism in Sri Lanka, where tourists often travel long distances on congested and winding roads, in vehicles driven by professional but sometimes fatigued drivers. Research on driver monitoring distinguishes between driver state (drowsiness, distraction, stress) and driving behavior (harsh braking, rapid acceleration, sharp cornering, speeding). Vision-based drowsiness detection commonly uses facial landmarks to compute the eye aspect ratio and detect prolonged eye closure, as proposed by Soukupova and Cech [21], together with yawning and head-pose cues. Behavior-based approaches use smartphone accelerometer and gyroscope signals, or vehicle data from the on-board diagnostics port, to detect harsh events and to score driving style.

Commercial telematics and fleet-management products exist, but they are typically designed for logistics fleets, require installed hardware, and report incidents after the fact in post-trip dashboards. Smartphone-based solutions reduce cost but often address a single risk (for example drowsiness alone), and they rarely combine driver state, vehicle condition and route risk into one proactive alert. They are also rarely connected to the planning and destination layers of a tourism system, so the safety information does not influence the itinerary or the choice of departure time. Component C3 addresses this by combining camera-based driver-fatigue monitoring and ultrasonic forward-collision sensing into a single Composite Safety Index on a low-cost in-vehicle IoT unit. It issues tiered proactive alerts to the driver and a minimal notification to the fleet administrator, in the context of a tourism trip.

### 2.2.4 Tourism flow prediction and optimization (C4)

Overcrowding at popular attractions is a recognized sustainability problem. Visitor-flow management has been approached through forecasting, simulation and optimization. Time-series models, including recurrent neural networks such as LSTM [19] and classical seasonal models, forecast arrivals at an attraction by hour, day and season using historical counts, calendars, weather and events. Pedestrian and crowd simulation, beginning with the social force model of Helbing and Molnar [22], explains how congestion forms in constrained spaces such as stairways and narrow passages.

Optimization approaches then redistribute demand through staggered entry slots, alternative routes, dynamic pricing or recommendations to visit less crowded sites.

Most published work addresses a single site or a single mechanism, relies on counting infrastructure such as turnstiles, cameras or Wi-Fi sensing that is expensive to deploy at open sites, and stops at prediction without delivering the recommendation to the tourist. Few studies join demand forecasts with the plans that tourists are actually making and with how they feel at the site. Component C4 addresses this by forecasting crowding across a network of attractions from a combination of historical data, bookings and live signals, and by generating redistribution recommendations that can be delivered through the planning component and the visitor application.

### 2.2.5 Integrated smart tourism platforms

Buhalis and Amaranggana [3] and Gretzel et al. [2] describe smart tourism destinations as ecosystems in which data from many sources are shared among stakeholders to enhance experiences and management efficiency. In practice, most deployed systems are single-purpose applications: a journey planner, a ticketing app, a navigation app or a survey tool. Data stay in separate silos and the stakeholders (tourists, drivers, site managers and authorities) do not share a common picture of what is happening. Integrating components requires agreed data contracts, consistent identity and consent handling, and a common spatial model of places, and it raises additional privacy and governance questions when sensitive signals such as emotion or driver state are involved. The review found few reports of tourism platforms that integrate planning, emotion sensing, transport safety and flow optimization in one architecture, particularly for a developing-country destination.

### 2.2.6 Summary of the reviewed literature

**Table 2.2.1: Summary of reviewed literature and gaps left open**

| Area / Study | What it does | Limitation relevant to this work |
|---|---|---|
| Conversational planning (BERT [6], NSGA-II [7]) | Intent extraction and multi-objective package optimization | Itinerary static after generation; no real-time adaptation or proactive voice support |
| Gonzalez-Rodriguez et al. [9] | Links facial expression analysis to tourist satisfaction | Laboratory stills; no real-time, mobile or spatial analysis |
| Kwon and Yu [11] | Real-time multi-CNN emotion recognition in a museum | Fixed indoor cameras; no mobile, zone mapping, trajectory or anomaly detection |
| Qi and Han [12] | CNN-based emotion detection in tourism emergencies | Reactive; no continuous monitoring |
| Matsuda et al. [13]; Santamaria-Granados et al. [14] | Wearable-based emotion estimation and recommendation | Specialized hardware; not scalable; no zone mapping |
| Khare et al. [15]; Kalateh et al. [16]; Ezzameli and Mahersia [17] | Reviews of multi-modal emotion recognition | Confirm the in-the-wild gap; tourism applications almost unexplored |
| Soukupova and Cech [21] | Eye-aspect-ratio drowsiness detection | Single risk factor; not connected to trip planning or route risk |
| Telematics and fleet systems | Driving behavior scoring and incident reporting | Installed hardware; post-trip reporting; not designed for tourism |
| Forecasting and crowd models (LSTM [19], social force [22]) | Arrival forecasting and congestion simulation | Single-site focus; no link to tourist plans or emotional feedback |
| Smart tourism literature [ [2], [3] | Conceptual frameworks for data-driven destinations | Few integrated implementations; privacy and governance of shared data under-specified |

### 2.2.7 Research gap

The literature review reveals a consistent, multi-dimensional gap. No reviewed study combines conversational and optimization-based trip planning, real-time smartphone-based emotion recognition at destinations, proactive driver and vehicle safety monitoring and crowd-flow optimization in one deployable, privacy-aware platform in which the components exchange information. Seven specific gaps are identified.

**Gap 1: Static, single-objective planning (C1).** Existing planners generate an itinerary once and rarely trade off cost, time, interest match and comfort explicitly. They do not understand multiple intents in one request and do not support voice-first, proactive interaction.

**Gap 2: No real-time adaptation of an itinerary (C1, C2, C4).** Itineraries do not respond to live conditions such as weather, closures, crowding or the emotional state of the travelers, so a plan that was good in the morning can become poor by the afternoon.

**Gap 3: No mobile platform for tourist emotion recognition (C2).** Every reviewed emotion-sensing system in tourism relies on fixed camera infrastructure [11] or wearable devices [13], [14]. None runs on the visitor's own smartphone, maps emotion to zones of a destination, models how emotion changes during a visit, or detects anomalous patterns before they become incidents.

**Gap 4: Fragmented, reactive transport safety (C3).** Driver-state monitoring and forward-collision warning are treated as separate functions with separate alerts, and are not linked to the tourist journey.

**Gap 5: Forecasting without redistribution (C4).** Crowd forecasts are seldom converted into recommendations that reach the tourist at the time of planning or during the visit, and the forecasts do not use emotional or planning signals.

**Gap 6: No integrated, closed-loop architecture.** The components that would make a smart tourism platform useful (planning, experience sensing, transport safety and flow management) are developed as separate silos, so that no component can use the outputs of another to improve its decisions.

**Gap 7: Privacy and governance of sensitive tourist data.** Emotion, location and driver-state data are sensitive. Few studies describe a design in which consent, pseudonymous identity, on-device processing and role-based access are consistent across all components and comply with the Personal Data Protection Act, No. 9 of 2022 of Sri Lanka [5].

### 2.2.8 Critical synthesis of the literature

The preceding review treated each component separately. This section draws the strands together and explains how the findings shaped the design decisions of the framework. Five themes recur across the literature and each leads to a specific design choice.

### 2.2.9 Theme 1: the gap between laboratory and field performance

Reviews of emotion recognition [15], [17] and of driver monitoring agree that systems that perform well on curated datasets lose accuracy when they are used outdoors, in vehicles, with spontaneous behavior and with diverse users. Illumination varies from deep shade to direct tropical sun, faces are partly covered by sunglasses, hats and masks, heads are turned toward monuments and not toward the camera, and cultural display rules mean that expressions are often more reserved than those in Western-collected datasets. The consequence for this project is twofold. First, targets for benchmark accuracy are set conservatively and are supplemented by measurements on pilot data, so that the claims of the work rest on field evidence rather than on benchmark figures alone. Second, the design uses quality gates, confidence thresholds, re-sampling and fallback behavior, so that an uncertain reading is discarded or down-weighted instead of being treated as a fact.

### 2.2.10 Theme 2: context improves interpretation

The strongest argument from the emotion literature is that a facial configuration has no fixed meaning [18], and the strongest empirical support is that fusion of several signals outperforms any single signal in naturalistic conditions [13], [16]. The same logic applies to the other components. A driver's eye closure is more meaningful when the hour, the length of the trip and the road type are known. A crowd count is more meaningful when the opening hours, the capacity of the zone and the plans of arriving tourists are known. A request such as "somewhere quiet" is more meaningful when the traveler's past choices and the current crowd forecast are known. This reasoning is the justification for the shared spatial model and the event interfaces: context is not an optional extra but the main means by which weak individual signals become useful.

### 2.2.11 Theme 3: time matters

Snapshot classifications describe a moment, whereas satisfaction, fatigue and crowding are processes. The emotion literature has largely focused on single frames, the driver-monitoring literature on thresholds applied to instantaneous values, and the crowd literature on counts at one time. Recurrent models [19] and sliding-window analysis offer a way to describe how a state develops, and the anomaly detection literature [20] distinguishes point anomalies from contextual and collective anomalies, the latter two of which are exactly what a sustained negative trajectory or a gradually rising occupancy represents. The framework therefore uses temporal models in C2, C3 and C4, and uses early-warning signals so that action can precede the incident.

### 2.2.12 Theme 4: from detection to decision

A recurring criticism [10], [8] is that technology studies in tourism stop at measurement. A heatmap that nobody acts on does not improve a visit. The framework therefore treats the recommendation as a first-class output and designs each component to close the loop: the planner offers a repaired itinerary, the emotion component offers a suggestion and a manager recommendation, the safety component offers a rest stop and the flow component offers an alternative time or site. The outcomes of the advice (accepted, declined, ignored) are recorded so that the effect of the recommendations can be evaluated rather than assumed.

### 2.2.13 Theme 5: privacy and trust determine whether the system can be used

Emotion, location and driver-state data are sensitive and, in several jurisdictions, regulated. The smart tourism literature [2], [3] notes that data-driven destinations depend on the trust of visitors and residents, and the Sri Lankan Personal Data Protection Act [5] sets obligations for consent, purpose limitation, data minimization and security. Acceptance studies in other domains show that users accept monitoring when they understand the purpose, when they control it and when they see a benefit. The design principles of on-device processing, granular consent, pseudonymous identity, aggregation for managers and transparent explanation follow directly from this observation and are not add-ons.

### 2.2.14 Implications for the framework

**Table 2.2.2: Literature themes and the resulting design decisions**

| Theme | Finding in the literature | Design decision in this framework |
|---|---|---|
| Laboratory-to-field gap | Benchmark accuracy overstates field accuracy [15], [17] | Conservative targets; pilot-based evaluation; quality gates; confidence thresholds; fallback behavior |
| Context improves interpretation | Facial configurations are ambiguous [18]; multi-modal beats uni-modal [13], [16] | Zone-wise emotion summaries in C2 |
| Time matters | Snapshots miss development of states; recurrent models suit sequences [19] | Emotion trajectory tracking over time (C2); consecutive-frame EAR fatigue trigger (C3): rolling-origin forecasting (C4) |
| Detection to decision | Few systems produce actionable recommendations [10], [8] | Recommendation engines in every component; outcomes of advice recorded |
| Privacy and trust | Acceptance depends on purpose, control and benefit [2], [3]; legal duties [5] | On-device processing; granular consent; pseudonymous identity; aggregation; audit logging |
| Single-site, single-purpose systems | Silos limit value [2], [3] | Integrated architecture with versioned contracts |

## 2.3 Subproblem and component mapping

Each of the four subproblems corresponds to one stage of the tourist journey and is owned by one component: planning by C1, the destination experience by C2, transport safety by C3, and crowd flow by C4. The integration problem is shared by all four. The subproblems have distinct core methods but are interdependent: C1 consumes crowd forecasts from C4 and emotion anomalies from C2, C3 supplies rest-stop requests to C1, and C2 and C4 exchange zone-level information. Each subproblem is necessary, because without it one stage of the journey would remain unsupported, and without integration the outputs of the components could not change one another's decisions.

## 2.4 Overall aim and group objectives

### 2.4.1 Main objective

To design, develop and validate an integrated, AI-based framework for intelligent travel support in Sri Lanka that unifies conversational trip planning with real-time adaptation, tourist emotion recognition, proactive driver and vehicle safety, and tourism flow optimization, delivered through mobile and web applications, and to evaluate its technical performance, usability and feasibility in a pilot.

### 2.4.2 Specific objectives

**Component C1 (AI conversational travel analyzer, package generation and adaptive engin**e)

- To build a multi-intent natural language analyzer that extracts destinations, dates, budget, group composition and interests from text and voice requests.
- To develop a route-ranking and package recommendation approach that consider multiple travel objectives and generates ranked, personalized travel packaged based on user preferences and trip constraints.
- To implement a real-time adaptive engine and a proactive voice agent that revise the itinerary and notify the traveler when conditions change.

**Component C2 (adaptive emotion recognition**)

- To develop an on-device facial emotion recognition module using a custom-trained convolutional neural network (CNN) that classifies five emotions and achieves the target accuracy on the RAF-DB datasets.
- To develop zone-wise satisfaction analysis using emotion records associated with each zone.
- To build an emotional trajectory tracker , a real-time anomaly detector with configurable thresholds, and a manager dashboard with zone heatmaps and recommendations.

**Component C3 (smart driver and vehicle proactive safety)**

- To train and evaluate lightweight MobileNetV2 models on public datasets (MRL Eye and YawDD) for on-device eye-state and yawn detection.
- To establish fatigue and collision-risk metrics using Eye Aspect Ratio (EAR) and Time-to-Collision (TTC), merging them into a Composite Safety Index (CSI).
- To develop a privacy-focused IoT architecture using an ESP32-CAM, an HC-SR04 ultrasonic sensor, an ESP32 main hub, and a SIM7600CE cellular module, ensuring no video data leaves the vehicle.
- To design a tiered alert mechanism (cabin buzzer, real-time driver app, and minimal FCM fleet notifications) that reserves steering wheel vibration strictly for the highest critical risk level.

**Component C4 (tourism flow optimization)**

- To build a crowd forecasting model for a network of attractions from historical, calendar, weather and live signals.
- To implement an optimization and recommendation engine that redistributes visitors through alternative sites, times and routes.
- To provide an operator dashboard for capacity monitoring, scenario analysis and recommendation tracking.

**Integration (common objectives)**

- To define a versioned data contract and event interfaces through which the four components exchange information.
- To implement consistent consent, pseudonymous session identity, security and access control across the framework.
- To demonstrate at least three cross-component scenarios end to end and to evaluate them in a pilot.

## 2.5 Scope, assumptions and constraints

The framework is a research prototype designed for adult volunteers who give informed consent and use their own smartphones. Emotion estimates are probabilistic indicators of experience and not psychological diagnoses. Driver monitoring is designed to support driver welfare and safety, not to discipline drivers, and its outputs are advisory. Crowd forecasts are limited to the attractions and the data sources available to the team. The pilot is limited to a small number of sites, vehicles and participants, so the evaluation demonstrates feasibility and effectiveness at pilot scale rather than national scale. The detailed methods, results and limitations of each component are reported in the corresponding individual dissertation, and this report focuses on the framework, the integration and the group-level evaluation. Where a statement in this report describes the internal design of components C1, C3 or C4, the individual dissertation is the authoritative source.

- Visitors have a smartphone with a front camera, GPS and sufficient processing power, and are willing to install the application.
- A usable face is visible for some part of each visit, for example while the visitor reads, takes a photograph or looks at the map.
- Site authorities are willing to provide a zone map, capacity figures and access for the pilot.
- Historical visitor counts for at least some sites can be obtained, or can be approximated from public data and the pilot.
- Drivers agree to the ESP32-CAM node being mounted in a position that gives the camera a clear view of the face, and the vehicle supplies power to the in-vehicle unit.
- The project budget and time are limited to one academic year, which restricts the scale of data collection and of the pilot.

# 3 OVERALL SOLUTION STRUCTURE

## 3.1 Integrated solution overview

The framework is one mobile-first system rather than four separate products. A visitor describes a trip in text or voice, and C1 turns the request into ranked packages and keeps the itinerary valid as conditions change. During the visit, C2 senses experience on the visitor's own device and returns suggestions to visitors and recommendations to managers. During road transport, C3 monitors driver fatigue and collision risk on a vehicle-mounted unit and issues tiered alerts. Across the destination network, C4 forecasts crowding and recommends redistribution. The central value is a closed loop in which a finding in one component informs the decisions of another, while raw sensor data are processed at source.

## 3.2 High level architecture

The framework is organized as a mobile-first, service-oriented system with four layers, shown in Figure 1 The **experience layer** consists of a tourist mobile application built with React Native, a driver mode within the same application, and a web dashboard for destination managers, operators and authorities. The **component layer** holds the four components C1 to C4, each of which exposes its functions as services and runs part of its processing on the device and part in the cloud. The **integration layer** provides the versioned REST API contract, an event bus for asynchronous messages, the pseudonymous session and consent services and the shared spatial model. The **data platform** holds the PostgreSQL and PostGIS database, the model registry and configuration, authentication and role-based access, and monitoring and audit logs.

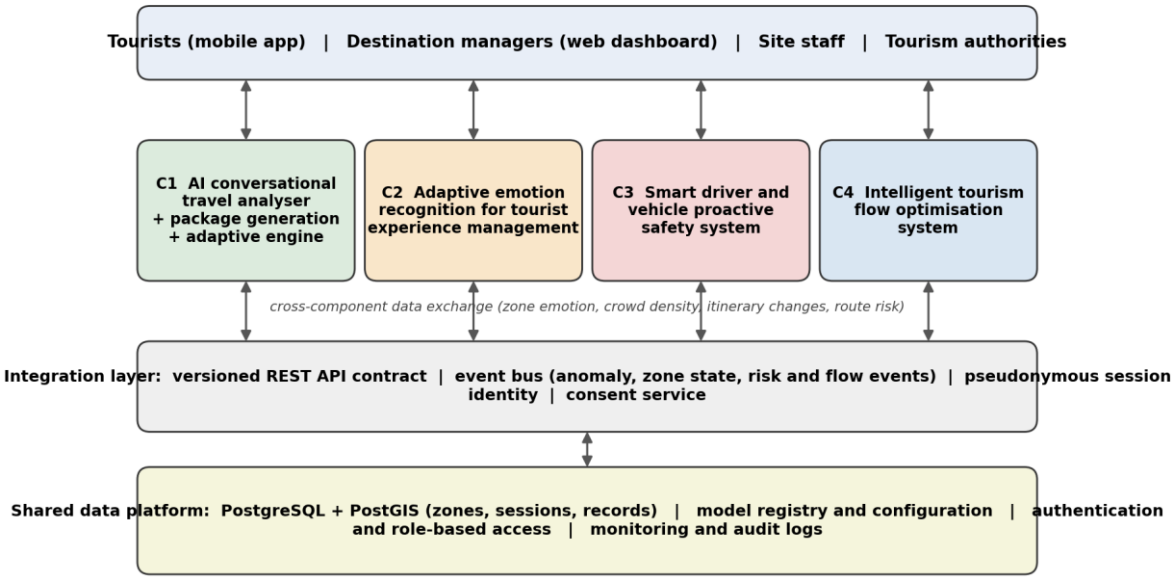



**Figure 3.3-1 : Integrated architecture of the R26-IT-112 framework**

Three design principles shaped this architecture. First, **independence**: each component can be deployed and evaluated on its own, and fails gracefully when another component is unavailable. Second, **privacy by design**: sensitive raw signals such as camera frames are processed on the device and are never transmitted, and only compact, pseudonymous records cross the network. Third, **explainability**: every alert and recommendation is traceable to the data and the rule or model that produced it, because managers and operators need to trust and audit the output.

## 3.3 Component design decisions

A design is only convincing if the alternatives that were rejected are known. This section records the main decisions behind the framework, the alternatives that were considered and the reasons for the choices, and then discusses stakeholders, scalability, legal and regulatory matters, accessibility and sustainability.

**Table 3.3.1: Principal design decisions and alternatives considered**

| Decision | Alternatives considered | Choice | Reason |
|---|---|---|---|
| Where to run emotion and driver-state inference | Cloud inference; edge server at the site; on-device | On-device | Privacy (no images leave the phone), scalability (no central bottleneck) and offline operation |
| Mobile platform | Separate native apps; progressive web app; cross-platform framework | React Native | One code base for Android and iOS with access to camera, location and native inference modules |
| Facial classifier backbone | Larger CNNs; pretrained CNN architectures | Custom CNN[4] | Trained on RAF-DB to classify five facial emotion classes using 100 × 100 grayscale inputs |
| Face pre-processing | Simple detector with crop; landmark-based alignment | OpenCV Haar Cascade | Detects the largest face, with −15° and +15° rotation fallback when initial detection fails. (C2) |
| Zone satisfaction mapping | Individual records; zone summaries | Zone-wise satisfaction analysis | Summarizes detected emotions by zone. |
| Emotion trajectory tracking | Individual readings; time-ordered history | Time-ordered emotion tracking | Displays emotion changes using recorded timestamps. |
| Anomaly detection | Statistical methods; supervised classifiers; configurable rules | Rule-based detection | Uses predefined rules to identify unusual patterns in recorded emotions. |
| Multi-objective planning | Weighted-sum optimization; NSGA-II; reinforcement learning | NSGA-II [7] | Provides a Pareto set that can be re-weighted by the user without re-running the search |

| Decision | Alternatives considered | Choice | Reason |
|---|---|---|---|
| Integration style | Shared database; direct API calls; event bus with versioned contracts | Event bus plus REST contracts | Loose coupling, independent deployment and a clear audit trail |
| Spatial representation | Raw coordinates; grid cells; semantic zones and segments | Semantic zones and segments | Meaningful to managers, supports privacy by design, and joins data across components |
| Back-end framework | Node.js; Django; FastAPI | FastAPI | Asynchronous, concise, and well suited to serving ML models |
| Database | Document store; relational; relational with spatial extension | PostgreSQL with PostGIS | Spatial queries on zone polygons and robust transactional guarantees |
| Recommendation logic | General place recommendations; interest-based recommendations | Interest-based recommendations | Suggests places based on users' selected interests. |

### 3.4 Component 1: Summary

Owner: Senevirathna T.M.N.D.L (IT22082442). Subproblem: planning and trip adaptation. Inputs: free-text and spoken travel requests, together with events from C2, C3 and C4. Core methods: a language model for multi-intent extraction, route-ranking and based selection for package generation, and a rule-based adaptive engine with a proactive voice agent. Outputs: ranked packages and repaired itineraries, published as ItineraryChanged events. Contribution: the decision layer through which experience, safety and crowd information reach the tourist's plan. Method detail is given in Appendix P and in the C1 dissertation.

### 3.5 Component 2 Summary

Owner: Kaweeshwara M.D.O (IT22216014). Subproblem: adaptive emotion recognition for tourist experience management. The component uses a custom CNN trained on RAF-DB to classify five facial emotion classes. OpenCV Haar Cascade is used for face detection, followed by grayscale preprocessing, normalization, and crop-ensemble inference. The component provides zone-wise satisfaction analysis, tracks changes in detected emotions over time, and recommends places based on users' selected interests. Method details are provided in Appendix P and the C2 dissertation.

### 3.6 Component 3 Summary

Owner: Rupasinghe M.L.K.S. (IT22176974). Subproblem: driver and vehicle proactive safety. Inputs: driver-face video processed on an ESP32-CAM and forward distance from an ultrasonic sensor. Core methods: on-device MobileNetV2 eye-state and yawn models, Eye Aspect Ratio, Time-to-Collision and the Composite Safety Index computed on an ESP32 hub. Outputs: tiered in-cabin and driver-app alerts, fleet notifications that carry no location, and DriverRiskAlert events. Contribution: the transport safety signal and rest-stop requests used by C1. Method detail is given in Appendix P and in the C3 dissertation.

### 3.7 Component 4 Summary

Owner: Manthrirathne M.A.A.C (IT22607614). Subproblem: intelligent tourism flow optimisation. Inputs: historical visitor counts, calendar and weather data, itineraries from C1 and live signals from C2. Core methods: a crowd score, an LSTM occupancy forecast with a seasonal baseline for comparison, and a redistribution optimiser. Outputs: CrowdForecast and RedistributionAdvice events and an operator dashboard. Contribution: the crowd forecasts used by C1 and the alternatives that change visitor plans. Method detail is given in Appendix P and in the C4 dissertation.

### 3.8 End to end user and data flow

The framework is evaluated through three end-to-end scenarios that require at least two components to cooperate. They are described here because they define what "integrated" means in this project and because they are the basis of the integration tests in Chapter 3.

***Scenario A: crowd-aware itinerary adaptation (C4 and C1)***

A tourist has an itinerary that includes a visit to a popular rock fortress at 10 a.m. The flow forecast of C4 predicts that the site will be at 95 percent of its comfortable capacity between 9 and 12 a.m. on that day because of a school holiday. C4 publishes a Redistribution Advice event. The adaptive engine of C1 evaluates the effect of moving the visit to 7 a.m. or swapping the order of two sites, checks the opening times, the travel times and the user's preferences, and the voice agent offers the revised plan. If the tourist accepts, C1 publishes Itinerary Changed, which C4 uses to update its demand estimate and C3 uses to update the expected departure time of the driver.

***Scenario B: emotion-triggered plan change (C2 and C1)***

During a visit, C2 detects a pattern of sustained negative emotion in a congested climbing section and sends a gentle suggestion to the visitor. If the same pattern appears for many visitors in the same zone, Rule C (zone dissatisfaction) fires and a ZoneEmotionState update is published. The adaptive engine in C1 sees that other travelers planned to visit this zone in the next hour and offers them an alternative order of attractions. The manager dashboard shows the zone in orange with the supporting evidence.

***Scenario C: fatigue-aware transport and visit plan (C3, C1 and C4)***

Late in a long day, the driver-state module of C3 detects repeated eye closure (low EAR), and the ultrasonic sensor detects a rapidly closing gap to the vehicle ahead (low TTC). The ESP32 hub fuses these into a critical Composite Safety Index (CSI) score. C3 issues a DriverRiskAlert that recommends a rest stop. C1 then proposes inserting a refreshment stop at a nearby site and shortening the final visit, and C4 checks that the proposed stop is not itself overcrowded. The tourist receives one coherent suggestion instead of three separate messages.

Table 21 gives the step-by-step sequence of Scenario A, which is representative of the others, and shows the messages exchanged and the component responsible for each step.

**Table 3.8.1: Message sequence for Scenario A (crowd-aware itinerary adaptation)**

| Step | Component | Action | Message |
|---|---|---|---|
| 1 | C4 | Daily forecast run predicts high occupancy at a site for the morning | CrowdForecast |
| 2 | C4 | Optimizer evaluates candidate actions and selects advice | RedistributionAdvice |
| 3 | C1 | Adaptive engine finds itineraries that include the site in the affected slot | - |
| 4 | C1 | Engine generates repair options respecting opening hours, travel time and preferences | - |
| 5 | C1 | Voice agent presents the options to the tourist | - |
| 6 | Tourist | Accepts one option by voice or touch | - |
| 7 | C1 | Itinerary is updated and the change is published | ItineraryChanged |
| 8 | C3 | Driver's expected departure time is updated | - |
| 9 | C4 | Demand estimate is updated and acceptance is recorded for evaluation | - |
| 10 | Shared | Audit entries are written for each cross-component flow | AuditEntry |

# 4 COMPONENT INTEGRATION AND COHESIVE OPERATION

## 4.1 Integration strategy

Integration was planned before the components were complete. Interface contracts were agreed early and kept in a shared contracts repository, every component was built to run stand-alone with a fallback, and the three cross-component scenarios were used as acceptance tests for the combined system. The comparison of integration styles that led to the event-bus design is given below.

Three ways of connecting the components were compared before the event-based approach was chosen. A shared database is simple but couples every component to one schema and makes independent deployment difficult, and it allows any component to read data it does not need, which is contrary to the privacy principle of data minimization. Direct point-to-point API calls are easy to understand, but with four components the number of connections grows and each component must know where the others are and whether they are available. An event bus with versioned contracts lets each component publish what it knows and subscribe to what it needs, tolerates the absence of a consumer, and provides a natural place to record an audit trail. Its cost is additional infrastructure and the need to design schemas carefully, which the project accepted.

**Table 4.1.1:Comparison of integration styles**

| Criterion | Shared database | Direct API calls | Event bus with contracts |
|---|---|---|---|
| Coupling between components | High | Medium | Low |
| Independent deployment | Difficult | Possible | Easy |
| Tolerance of an unavailable component | Low | Low | High |
| Data minimization between components | Poor | Good | Good |
| Auditability of data flows | Poor | Moderate | Good |
| Infrastructure complexity | Low | Low | Medium |
| Suitability for real-time notification | Poor | Good | Good |

## 4.2 Interfaces, data contracts and shared models

All components describe place in the same way. The spatial model has three levels. A **site** is a destination such as a heritage complex or a national park. A **zone** is a named, bounded area within a site, with a semantic type such as entrance pathway, main monument, climbing section, viewing platform, rest area or exit, and with a set of amenities and a capacity. A **route segment** is a section of road between two destinations, with attributes such as length, typical travel time, road class and historical risk. Zone polygons and route geometries are stored in PostGIS, and every record in the framework that refers to place carries a zone or segment identifier rather than raw coordinates. This allows an emotion record from C2, a crowd forecast from C4 and a route-risk score from C3 to be joined on identifiers, and also supports the privacy rule that coordinates are converted to identifiers on the device.

A session is opened with a consent receipt that lists the services the user has agreed to, for example emotion sensing, driver monitoring and location-based suggestions, and the version of the consent text. The service returns a short-lived token scoped to that session and a random pseudonymous identifier. The identifier rotates between sessions and is not linked to a name, a phone number or a device identifier. Each component checks the consent receipt before it processes or stores data for a given purpose, so that a visitor who has agreed to itinerary assistance but not to emotion sensing does not have the camera used for emotion. A user can withdraw consent at any time, which stops further collection and triggers deletion according to the retention policy.

Components exchange information through two mechanisms. **Request and response interfaces** are versioned REST endpoints described in an OpenAPI specification. **Events** are published on an event bus when something happens that other components might use. Every message carries a schema version, a message type, a timestamp, the pseudonymous session or an aggregate scope, and the site, zone or segment identifier. Receivers validate each message against its schema and reject malformed or out-of-range values. The principal events are listed in Table 5.

**Table 4.2.1: Principal cross-component events and interfaces**

| Event / interface | Producer | Consumers | Content |
|---|---|---|---|
| ZoneEmotionState | C2 | C1, C4, dashboard | Aggregated satisfaction, valence and sample count per zone and time window |
| EmotionAnomaly | C2 | C1, site staff | Anomaly type, severity, zone, evidence reference |
| CrowdForecast | C4 | C1, C2 | Predicted occupancy by site, zone and hour with an uncertainty range |
| RedistributionAdvice | C4 | C1, tourist app | Recommended alternative site, time or route and the reason |
| ItineraryChanged | C1 | C3, C4 | New plan: sites, times and expected departures |
| DriverRiskAlert | C3 | Driver, operator, C1 | Alert level, alert type and timestamp, plus an optional rest-stop recommendation for C1. The FCM copy sent to the fleet administrator carries only the alert type and timestamp (no GPS or video). |
| TripSafetySummary | C3 | C1, operator | End-of-trip safety score and the main events |
| ZoneConfiguration | Dashboard | C2, C4 | Thresholds, capacities and no-capture zones |

Table 17 lists the principal REST endpoints of the framework. All endpoints are versioned under a prefix such as /v1, require a valid short-lived token except for session creation, and validate request bodies against the schemas in the contracts repository. Requests that fail validation receive a 422 response with a list of the offending fields, and idempotent writes use the message identifier so that retries are safe.

**Table 4.2.2: Principal REST endpoints**

| Method and path | Component | Purpose | Main inputs and outputs |
|---|---|---|---|
| POST /v1/sessions | Shared | Open a session with a consent receipt | In: consent receipt, selected services. Out: token, pseudonym, site configuration |
| DELETE /v1/sessions/{id} | Shared | Withdraw consent and close the session | Out: deletion confirmation |
| GET /v1/sites/{id}/zones | Shared | Download the zone model and configuration | Out: zone polygons, types, no-capture flags, thresholds |
| POST /v1/chat/messages | C1 | Send a text request to the analyser | In: message, locale. Out: intents, entities, reply, follow-up question |
| POST /v1/chat/voice | C1 | Send transcribed speech | In: transcript, confidence. Out: as above |
| POST /v1/packages/generate | C1 | Generate ranked packages | In: constraints and weights. Out: packages with objective scores |
| POST /v1/itineraries/{id}/events | C1 | Notify the adaptive engine of an event | In: event type and details. Out: repair options |
| POST /v1/emotion/records | C2 | Upload a batch of emotional state records | In: pseudonym, zone ID, timestamp, probabilities, valence, confidence. Out: accepted count, any suggestions |
| GET /v1/emotion/zones/{id}/state | C2 | Read aggregate emotion state of a zone | Out: satisfaction, valence trend, sample count |
| GET /v1/emotion/anomalies | C2 | List anomalies with evidence | In: site, time range, type. Out: anomaly list |
| POST /v1/driver/trips | C3 | Start or close a monitored trip | In: calibration summary. Out: trip identifier |

| Method and path | Component | Purpose | Main inputs and outputs |
|---|---|---|---|
| POST /v1/driver/trips/{id}/events | C3 | Upload driver-state and driving events | In: event batch. Out: alerts to display |
| GET /v1/flow/forecast | C4 | Read occupancy forecasts | In: site, horizon. Out: forecast with intervals |
| GET /v1/flow/advice | C4 | Read redistribution advice | In: site, date. Out: ranked advice items with reasons |
| POST /v1/flow/scenarios | C4 | Run a what-if simulation | In: acceptance rate or capacity change. Out: projected occupancy |
| PUT /v1/config/zones/{id} | Shared | Update thresholds, capacity and no-capture zones | In: configuration. Out: new version |
| GET /v1/audit/flows | Shared | Read the audit log of cross-component data flows | In: time range. Out: log entries (restricted to administrators) |

**Table 4.2.3: Principal entities of the data model**

| Entity | Key attributes | Relationships and notes |
|---|---|---|
| Session | pseudonym, start and end time, consent version, consented services | Parent of emotion records and trips; no personal identifiers |
| Site | identifier, name, opening hours, capacity, configuration version | Contains zones and is linked to route segments |
| Zone | identifier, site, type, polygon, amenities, capacity, no-capture flag | Referenced by emotion records and forecasts |
| RouteSegment | identifier, geometry, road class, length, curvature, gradient | Referenced by trip events and risk scores |
| CNN emotion probabilities | session, zone, timestamp, fused probabilities, valence, confidence, low-confidence flag | Used for trajectories and aggregates; idempotent by message ID |

| Entity | Key attributes | Relationships and notes |
|---|---|---|
| Anomaly | type, severity, zone, evidence reference, status, timestamps | Produced by rule-based checks; linked to alerts. |
| Recommendation | target (tourist or manager), text, rationale, status, outcome | Produced by decision support in C2 and C4 |
| Package and Itinerary | constraints, objective scores, selected sites, times, version history | Revised by the adaptive engine; versions retained |
| TripSummary | driver pseudonym, duration, distance, event counts, scores | Raw traces purged after scoring |
| Forecast | site, zone, hour, expected occupancy, interval, model version | Compared with actuals for evaluation |
| Advice | site, date, action, reason, projected effect, acceptance outcome | Closes the loop for C4 evaluation |
| AuditEntry | time, producer, consumer, message type, scope | Immutable; supports governance |

## 4.3 Shared nonfunctional requirements

Requirements were derived from the stakeholder analysis and the research objectives. The principal stakeholders are tourists, drivers, destination managers, site staff, transport operators and tourism authorities. Tourists use the app and receive suggestions. Drivers use driver mode and receive alerts. Managers monitor satisfaction and crowding and configure thresholds. Site staff receive safety alerts. Operators monitor trips and vehicles. Authorities use aggregated, anonymized data for planning. The group-level functional requirements are listed in Table 6 and the non-functional requirements in Table 7. Component-specific requirements are given in the individual dissertations.

**Table 4.3.1: Group-level functional requirements**

| ID | Requirement | Component | Priority |
|---|---|---|---|
| FR1 | The system shall accept travel requests in text and voice and extract multiple intents and constraints. | C1 | High |

| ID | Requirement | Component | Priority |
|---|---|---|---|
| FR2 | The system shall generate ranked, personalized packages that balance cost, time, interest match and comfort. | C1 | High |
| FR3 | The system shall revise an itinerary in real time when conditions change and notify the traveler. | C1 | High |
| FR4 | The app shall classify tourist facial emotion on the device and link it to the current zone. | C2 | High |
| FR5 | The system shall track emotional trajectories and detect anomalies, and send suggestions and alerts. | C2 | High |
| FR6 | The dashboard shall display a live zone emotion heatmap and generate zone recommendations. | C2 | High |
| FR7 | The ESP32-CAM shall detect driver fatigue (EAR/yawning) on-device without transmitting video. | C3 | High |
| FR8 | The HC-SR04 ultrasonic sensor shall measure forward distance to compute Time-to-Collision (TTC). | C3 | High |
| FR9 | The system shall fuse EAR and TTC into a Composite Safety Index (CSI) to trigger tiered proactive alerts. | C3 | Medium |
| FR10 | The system shall forecast crowd levels for each site and hour. | C4 | High |
| FR11 | The system shall recommend redistribution of visitors by site, time or route. | C4 | High |
| FR12 | The operator dashboard shall show capacity status, forecasts and recommendation outcomes. | C4 | Medium |
| FR13 | The framework shall exchange the events in Table 2.2 and react to them as described in Section 2.3.4. | All | High |
| FR14 | The framework shall obtain, record and honor user consent per service and allow withdrawal. | All | High |
| FR15 | The app shall provide reduced functionality offline and synchronize when connectivity returns. | All | Medium |
| FR16 | Managers and operators shall be able to configure thresholds, capacities and no-capture zones. | All | Low |

**Table 4.3.2: Group-level non-functional requirements**

| Category | Requirement |
|---|---|
| Performance | On-device inference shall complete within 150 ms per frame on a mid-range smartphone; interactive API calls shall respond within 2 seconds at the 95th percentile. |
| Scalability | The cloud back end shall support at least 500 concurrent active sessions, with each component scalable independently. |
| Privacy | No raw facial images or camera frames shall be stored or transmitted; coordinates shall be converted to zone or segment identifiers on the device, except that the C3 vehicle position is shown only on the driver's own live map and is never included in fleet notifications; data collection shall follow the Personal Data Protection Act, No. 9 of 2022 [5]. |
| Security | All app, cloud and dashboard communication shall use TLS 1.3, and the C3 in-vehicle ESP-NOW links shall use CCMP-protected frames; tokens shall be short-lived and session-scoped; dashboards shall use individual accounts with role-based access control. |
| Availability | Back-end availability of at least 99 percent during site opening hours; each component shall degrade gracefully if another is unavailable. |
| Usability | The app and dashboards shall achieve a System Usability Scale score of at least 70 [24]. |
| Battery | Active use of the app shall not consume more than 15 percent of battery capacity per hour. |
| Maintainability | Interfaces shall be versioned and documented; models shall be versioned with the data version and evaluation results. |
| Auditability | Every alert and recommendation shall record the evidence and the component that produced it. |

The framework handles several categories of data with different sensitivity. Table 8 lists them together with where they are processed, what is stored and how long it is retained. The general rule is that the most sensitive data stay on the device, and that what leaves the device is the minimum needed for the purpose.

**Table 4.3.3: Trajectory analysis and anomaly detection flow in component C2**

| Data | Processed | Stored / transmitted | Retention |
|---|---|---|---|
| Camera frames (faces) | On device, in memory only | Never stored or transmitted | Discarded immediately after inference |
| Emotion probability vectors | On device (fusion), cloud (trajectory) | Probability vector, zone ID, timestamp, session pseudonym | At most 12 months |
| Location | On device | Converted to zone or segment ID; coordinates not transmitted. For C3, the vehicle position is sent to Firebase only for the driver's live map and is never included in FCM alerts | Identifier only, as above |
| Driver fatigue and distance indicators | On device (ESP32-CAM and ESP32 hub) | Numerical status (EAR, TTC, CSI) and timestamped alerts | At most 12 months |
| Fleet administrator notifications | Cloud (FCM) | Alert type and timestamp only (no GPS or video) | Kept only until delivered |
| Travel requests and preferences | Cloud (C1) | Extracted entities and conversation text, linked to the session pseudonym | At most 12 months, deletable on request |
| Aggregated zone and site statistics | Cloud | Aggregated with sample counts; no individual identification | Longer for research, in anonymized form |

### 4.4 Integrated workflow and exception handling

Integration creates the risk that several components send contradictory or redundant messages. The framework therefore applies simple rules. Safety-related messages take priority over comfort and convenience messages. Messages to the same user are de-duplicated and rate-limited, and are bundled when they relate to the same decision, as in Scenario C. A recommendation that has been declined by the user is not repeated for a configurable period. Every message that reaches a user records which component and which evidence produced it, so that the origin of a suggestion can be audited.

All communication uses TLS 1.3. Tokens are short-lived and scoped to a single session. Dashboard users authenticate with individual accounts and are assigned roles such as manager, staff member, operator or analyst, which determine what they can see and change. Because personally identifying information is not collected, the attack surface is narrower and the consequence of any breach is smaller. Server logs exclude payload contents, secrets are held in a secret manager and rotated, and database backups are encrypted.

The design anticipates failures that are likely at outdoor sites and on roads. If the network is unavailable, records are queued and uploaded later in time order. If GPS is lost, the app keeps the last known zone for a limited time and then marks records as zone-unknown. If the camera is blocked or a face is not visible, the app produces no record rather than guessing. If a component receives no events from another, it falls back to its stand-alone behavior. If the back end is overloaded, clients reduce their upload rate and the server sheds low-priority work before alerting.

### 4.5 Team contribution and dependency matrix

The four students each own one component and are responsible for its design, implementation, evaluation and individual dissertation. The team met weekly to review progress, to resolve interface questions and to plan integration work. A single repository organization with one repository per component plus a shared contracts repository was used for version control. The contracts repository holds the API specification, the event schemas and the zone model, so that a change to an interface is visible to every member and is reviewed by the owners of the components affected.

Work was tracked on a shared task board, with milestones aligned to the progress presentations. Decisions that affected more than one component, for example the choice of identifier format or the retention period for records, were recorded in a decision log with the date, the options considered and the reason for the choice.

The project was carried out by a team of four students, each responsible for one component, with shared responsibility for the integration layer. Table 34 summarizes the contribution of each team member. The common report describes the integrated framework, while each individual dissertation covers the component developed by that student.

**Table 4.5.1: Summary of each student's contribution**

| Student (ID) | Component | Main individual contribution | Contribution to the integration |
|---|---|---|---|
| Senevirathne T.M.N.D.L (IT22082442) | C1: AI conversational travel analyzer with package generation and adaptive engine | Multi-intent NLP analyzer, route-ranking and rule-based package selection, real-time adaptive engine and proactive voice agent | Consumes events from C2, C3, and C4 and publishes ItineraryChanged events; contributed to the shared API contracts, event-driven integration, cross-component itinerary adaptation, and end-to-end integration testing. |

| Student (ID) | Component | Main individual contribution | Contribution to the integration |
|---|---|---|---|
| Kaweeshwara M.D.O (IT22216014) | C2: Adaptive emotion recognition for tourist experience management | Developed a custom CNN-based facial emotion recognition module using a five-class subset of RAF-DB. Implemented Haar Cascade face detection, grayscale preprocessing, normalization, and crop-ensemble inference. Developed zone-wise satisfaction analysis, emotion trajectory tracking, and interest-based place recommendations. | Provides zone-wise satisfaction summaries and emotion trajectory data for the manager dashboard. |
| Rupasinghe M.L.K.S. (IT22176974) | C3: Smart driver and vehicle proactive safety system | Designed the IoT edge architecture (ESP32-CAM, HC-SR04, SIM7600CE); trained MobileNetV2 eye-state and yawn models; formulated the Composite Safety Index (CSI); implemented Firebase real-time alerts and privacy-first FCM notifications. | Publishes DriverRiskAlert. The driver app shows the vehicle position, supplied by the SIM7600CE module, on a live map; the position is never included in fleet notifications. |
| Manthrirathne M.A.A.C (IT22607614) | C4: Intelligent tourism flow optimization system | Crowd forecasting, optimization and recommendation engine, operator dashboard | Publishes CrowdForecast and Redistribution Advice |

**Shared work.** The project was developed collaboratively through a shared contracts repository, common API gateway and event-bus architecture, consent and session services, and a shared database schema supporting communication between the four

components. The team jointly developed the common application shell and integration interfaces, conducted cross-component integration and resilience testing, and coordinated the pilot deployment and data collection activities. The members also contributed jointly to the preparation, review, formatting, and consolidation of this common report. Each member was primarily responsible for the design, implementation, evaluation, and documentation of their assigned component, while the shared infrastructure, integration activities, testing, pilot organization, and final report preparation were completed collaboratively across the team.

# 5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

## 5.1 Common technology stack and infrastructure

**Table 5.1.1: Tools and platforms used**

| Tool or platform | Purpose | Justification |
|---|---|---|
| React Native | Tourist and driver mobile application | Single code base for Android and iOS |
| TensorFlow and TensorFlow Lite | Model training and on-device inference | Mature mobile runtime with quantization support; works offline |
| TensorFlow / Keras - Custom CNN (C2) | Training and evaluation of the five-class facial emotion classifier using RAF-DB | Supports CNN development and classification using 100 × 100 grayscale inputs. |
| OpenCV - Haar Cascade (C2) | Face detection and preprocessing | Supports largest-face selection, rotation fallback, resizing and grayscale conversion. |
| MobileNetV2 (C3) | eye-state and yawn classification on the ESP32-CAM (C3); face landmarks for C2 | Lightweight architecture for eye-state and yawn classification. |
| ESP32-CAM, ESP32, HC-SR04, SIM7600CE, ESP-NOW, Firebase Realtime Database, FCM | C3 in-vehicle unit, data path and fleet notifications | Low-cost commodity modules; keeps video on the vehicle; no separate inference server |
| YOLOv8 nano | Crowd density estimate | Lightweight real-time person detection |
| Transformer language models | Intent and entity extraction; dialogue (C1) | Strong accuracy on multi-intent language understanding |
| NSGA-II implementation | Multi-objective package optimization (C1) | Produces Pareto-optimal alternatives |

| Tool or platform | Purpose | Justification |
|---|---|---|
| Python and FastAPI | Back-end REST services and model services | Concise, asynchronous, well-suited to ML services |
| PostgreSQL with PostGIS | Sessions, records and spatial zone storage | Spatial queries and polygon zones |
| Message broker (for example Redis streams or RabbitMQ) | Event bus between components | Decouples components; supports retries |
| React and Leaflet | Manager and operator dashboards with maps | Component-based UI with interactive maps and heatmaps |
| | | |
| Docker and a cloud provider | Packaging, hosting and GPU retraining | Independent scaling of each service |
| Git and a shared task board | Version control and planning | Traceable progress and team coordination |

## 5.2 Development and integration process

Development followed a lightweight agile process with two-week sprints. Each sprint began with planning against the user stories and ended with a demonstration to the supervisors. Code was reviewed by at least one other team member before it was merged, and continuous integration ran the unit and contract tests on every change. Models were trained from scripts under version control with fixed random seeds, and each trained model was stored in the registry with its data version, configuration and evaluation results. Defects found in testing were recorded in the task board with severity and were retested after correction. Before each progress presentation a release candidate was built and tested on the reference devices so that demonstrations used a known version of the system.

Documentation was treated as part of the deliverable. The contracts repository includes the interface specification and examples, each component repository includes setup

instructions and a description of its configuration, and the decision log records choices that affect more than one component. This allows a new developer, or an examiner, to rebuild the system and to understand why it was built as it was.

The work was divided into six work packages: research and design; data and model development; mobile application development; the intelligence and integration layer; testing and validation; and documentation and communication. Table 11 gives the schedule across the two semesters.

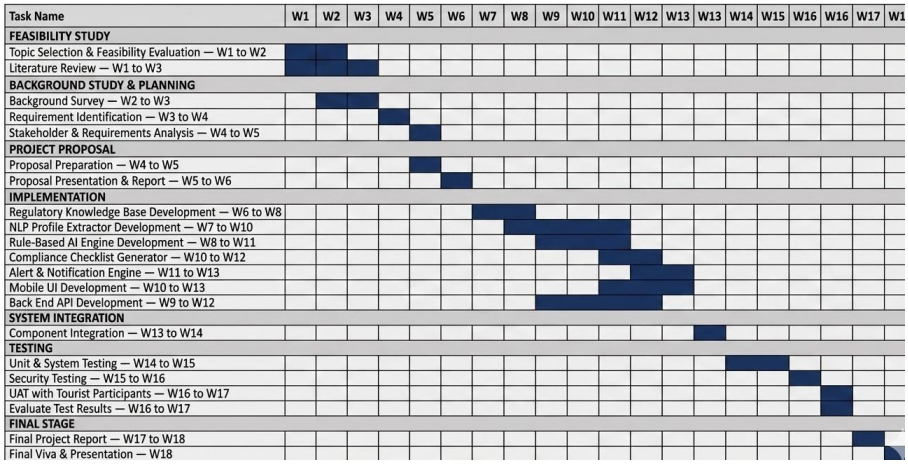



**Figure 5.5-1 : Work breakdown structure of the project**

**Table 5.2.1: Project schedule and milestones**

| Period | Activities | Milestone or output |
|---|---|---|
| Semester 1, Jan to Mar 2026 | Literature review, gap analysis, requirements, architecture and contract design | Project proposal presentation (March 2026) |
| Semester 1, Apr to May 2026 | Dataset preparation, first models, first app prototype, first API version | Progress presentation 1 (May 2026) |
| Semester 2, Jun to Aug 2026 | On-device integration, fusion, trajectory and anomaly modules, optimizer, forecasting, driver mode, dashboards, event bus | Progress presentation 2 (August 2026) |

| Period | Activities | Milestone or output |
|---|---|---|
| Semester 2, Sep to Oct 2026 | System integration, cross-component scenarios, testing, pilot, evaluation, dissertations and common report | Final presentation and submission (October 2026) |

**Table 5.2.2: Principal risks and mitigation**

| Risk | Likelihood | Impact | Mitigation |
|---|---|---|---|
| Low accuracy in outdoor lighting (C2, C3) | High | High | Adaptive pre-processing; outdoor augmentation; quality gate; confidence-based re-sampling |
| Integration failures between components | Medium | High | Early contract agreement; contract tests; stand-alone fallback in each component |
| Privacy concerns reduce consent rate | Medium | High | Granular consent; demonstration that no images are stored; opt-out at any time |
| Dataset bias (limited Asian faces) | High | Medium | Use RAF-DB; per-group metrics; field calibration |
| Limited face visibility while walking (C2) | High | Medium | Burst sampling; coverage metrics; limitation stated |
| Limited historical visitor data for forecasting (C4) | High | Medium | Transfer from similar sites; wider uncertainty; ongoing retraining |
| Camera mounting and lighting inside vehicles (C3) | Medium | Medium | Mounting guidance; per-driver calibration; infrared-friendly thresholds |
| Third-party service cost or limits (language model, maps) | Medium | Medium | Caching; rate limits; open alternatives; budget monitoring |
| Ethics approval delays | Medium | High | Early submission; fallback to laboratory, simulated and benchmark evaluation |
| GPS inaccuracy near zone boundaries | Medium | Medium | Hysteresis on zone changes; zone buffers; smoothing |
| Battery drain | Medium | Medium | Burst sampling; quantized models; duty cycling |
| Team member unavailability | Low | High | Shared repositories and documentation; cross-training on interfaces |

## 5.3 Deployment and operational configuration

On the device, the application runs the models that process raw sensor data: the facial emotion model and the face-quality checks (C2), the custom ESP32-CAM edge unit handles the driver-state landmark processing and models on-device, while the ESP32 main hub computes the Composite Safety Index (CSI) (C3) and the voice front end and light-weight intent handling (C1). The cloud runs the services that need aggregated data or heavy computation: the package optimizer and language-model services (C1), trajectory analysis, anomaly detection and zone aggregation (C2), the cloud exclusively handles Firebase Realtime Database synchronization and FCM push notifications, rather than heavy inference (C3), and forecasting and optimization (C4). The back end is a set of FastAPI services behind an API gateway, containerized so that each component can be scaled independently. Records of a few hundred bytes are sent every few seconds by each active session, so network load grows with the number of active users but processing per user stays small.

The application supports limited offline operation, which is necessary at remote heritage sites and on rural roads. When the network is unavailable, on-device inference continues, records are queued locally in time order and uploaded when connectivity returns. The server accepts records idempotently so that retransmissions do not create duplicates.

Behavior that managers or developers may need to tune is held in configuration and not in code. Table 19 lists the main parameters with their default values and where they are set.

**Table 5.3.1: Principal configuration parameters and defaults**

| Parameter | Default | Component | Set by |
|---|---|---|---|
| Face confidence threshold for re-sampling | 0.75 | C2 | Developer |
| Minimum interval between records | A few seconds | C2 | Developer |
| Valence smoothing window | Median over a short window | C2 | Developer |
| Rule A duration | 3 minutes at valence <= -1 | C2 | Manager |

| Parameter | Default | Component | Set by |
|---|---|---|---|
| Rule B drop | 2 levels within 30 seconds | C2 | Manager |
| Rule C baseline | Manager-defined or learned from history | C2 | Manager |
| Alert rate limit per session and zone | Configurable | C2, shared | Manager |
| No-capture zones | None until defined | C2, C3 | Manager |
| Zone hysteresis and buffer | Metres, per zone type | Shared | Developer |
| EAR threshold and consecutive frames | Calibrated per driver (EAR_thresh, N_frames) | C3 | Developer |
| Alert levels | Information, caution, urgent | C3 | Operator |
| Comfortable occupancy threshold | Fraction of capacity per site | C4 | Operator |
| Forecast horizon | 24 hours (extendable to days) | C4 | Developer |
| Advice cool-down after decline | Configurable period | Shared | Operator |
| Data retention | 12 months maximum | Shared | Administrator |
| Upload batch size and period | Small batches every few seconds | Shared | Developer |

## 5.4 Evidence of team execution

Evidence of execution comes from the shared task board, the Git history of the component and contracts repositories, the regular team meetings, the decision log that recorded choices affecting more than one component, and the sprint demonstrations to the supervisors. These records are summarised here rather than reproduced, and the raw activity logs are held by the team.

# 6 INTEGRATED TESTING EVALUATION AND VALIDATION

## 6.1 Evaluation framework and success criteria

The project follows the Design Science Research (DSR) paradigm [25]. DSR is appropriate when the main outcome of the research is a new technological artefact that addresses a real-world problem and is rigorously evaluated. Here the artefact is an integrated, AI-based travel support framework. The work is organized using the six activities of the DSR methodology of Peffers et al. [26], mapped to project phases in Table 4. Because the framework has four components developed by four students, the process was coordinated so that each component followed the same activities on the same schedule, and so that the integration contract was agreed early enough to be used by all components.

**Table 6.1.1: Mapping of DSR activities to project phases**

| DSR activity | Project phase | Main outputs |
|---|---|---|
| 1. Problem identification and motivation | Literature review and gap analysis (individual and joint) | Research gaps and research problem (Chapter 1) |
| 2. Definition of objectives | Requirements and stakeholder analysis | Objectives; functional and non-functional requirements (Section 2.4) |
| 3. Design and development | Component development; integration layer; applications | Working artefact (Sections 2.2, 2.3 and 2.5) |
| 4. Demonstration | Prototype demonstrations and progress presentations | Demonstrable prototype on real devices; cross-component scenarios |
| 5. Evaluation | Component benchmarking, integration testing, usability study, pilot | Quantitative and qualitative evidence (Chapter 3) |
| 6. Communication | Dissertations, common report, presentations | This report and the four dissertations |

Parts of the work are experimental as well as design-oriented, and several hypotheses are tested. These include that fusing context with facial probabilities improves emotion accuracy over a face-only baseline (C2), that the optimizer produces packages that dominate a baseline planner on cost, time and interest match (C1), that fusing EAR-based fatigue and TTC-based collision risk into a Composite Safety Index (CSI)

triggers accurate and timely tiered alerts (C3) that the forecasting model outperforms seasonal baselines (C4), and that cross-component information flow improves outcomes compared with the same components operating independently (integration).

**Table 6.1.2: Project schedule and milestones**

| Evaluation question | Data | Metrics and method |
|---|---|---|
| EQ1 (C1). Does the analyzer understand multi-intent requests and does the optimizer produce better packages than a baseline? | Held-out request set; baseline planner | Intent accuracy, slot F1; hypervolume, cost, time and interest-match against the baseline |
| EQ2 (C2). How accurately does the custom CNN classify five facial emotion classes? | Five-class subset of the RAF-DB test set | Accuracy, per-class precision, recall, F1-score, macro F1 and confusion matrix. |
| EQ3 (C2). Does the component correctly display zone-wise satisfaction and emotion trajectories, and provide recommendations matching users' selected interests? | Timestamped emotion records with zone identifiers; selected user interests; recommendation outputs | Functional checks of zone grouping, chronological ordering of emotion records, and matching recommended place categories to selected interests. |
| EQ4 (C3). Does the system detect driver fatigue and forward-collision risk and issue tiered alerts with few false alarms? | Recorded or simulated driver data; distance readings. | Precision, recall, F1, false alarms per hour; alert lead time |
| EQ5 (C4). Are crowd forecasts accurate, and does the advice reduce peaks? | Historical counts; pilot data; simulation | MAE, MAPE, interval coverage; simulated and observed peak reduction |
| EQ6 (integration). Do the components work together as designed and does integration add value? | Cross-component scenarios A to C | Scenario success rate, end-to-end latency, comparison with independent operation |
| EQ7 (system). Is the framework fast, efficient and usable? | Device and load tests; participants | Latency (median, 95th percentile), battery per hour, concurrency, SUS, Likert ratings |

**Table 6.1.3: Design targets**

| Measure | Target |
|---|---|
| Custom CNN test accuracy on the five-class RAF-DB subset | At least 80 percent |
| Per-frame on-device latency (mid-range phone, end to end) | Under 150 ms |
| Fatigue and collision-risk alert detection F1 (C3) | At least 0.80 |
| Crowd forecast MAPE (C4) | At most 15 percent (24-hour horizon) |
| Intent accuracy (C1) | At least 90 percent |
| Cross-component scenarios completed end to end | All three scenarios (A, B and C) |
| SUS score (applications and dashboards) | At least 70 |
| Battery consumption in active use | At most 15 percent per hour |
| Concurrent sessions supported | At least 500 |

Testing is organized in layers so that each module is verified in isolation before the whole system is evaluated.

- **Unit testing** of pre-processing functions, rule evaluation, entity extraction, objective functions, event detectors, zone lookup and API handlers using automated test suites.
- **Model evaluation** on held-out test sets for each learned model, reporting the metrics appropriate to the task: accuracy, precision, recall, macro and weighted F1 and confusion matrices (C2, C3 drowsiness), intent accuracy and slot F1 (C1), hypervolume and package quality (C1), and forecast error and interval coverage (C4).
- **Contract testing** to verify that each producer emits events that conform to the published schema and that each consumer handles valid, missing and out-of-range fields correctly.
- **Integration testing** of the three cross-component scenarios in Section 2.3.4, including component outages, the offline-to-online transition, GPS loss and low-connectivity behavior.

- **Performance testing** of per-frame latency and battery drain on at least two mid-range Android phones and one iOS phone, and load testing of the back end with simulated sessions up to and above 500 concurrent sessions.
- **Scenario-based safety and anomaly testing** using scripted and replayed trajectories and driving traces that include each anomaly type and normal variation, measuring precision, recall and detection delay.
- **Security and privacy testing**, including verification that no image data appears in network traffic or storage, checks of consent enforcement, access control and encryption.
- **Usability testing** of the application and dashboards using the System Usability Scale [24] and short questionnaires on perceived usefulness and trust.

Pilot data are stored in a separate, access-controlled schema and exported only in aggregated or anonymized form. Each export is documented with its date, query and purpose. Annotation of trajectories is carried out by two annotators who first receive written guidelines with examples and label a calibration set together, then label the remaining sessions independently without access to model predictions. Cohen's kappa is reported. To support reproducibility, random seeds, library versions, dataset splits and configuration files are recorded. Benchmark datasets follow the official split where one exists, the test partition is used only for final evaluation, and hyperparameters are tuned on the validation partition only. For field data, a fixed proportion of sessions is held out before model tuning, and sessions from the same participant are never split between training and test partitions.

So that the results in Section 3.1 can be interpreted and reproduced, this section defines the metrics, describes the procedure for each experiment, states the statistical analysis plan and sets out the rules for data quality and reporting. It is written as a protocol and is to be followed when the placeholders in Section 3.1 are completed.

### 6.1.1 Metric definitions

For each class, **precision** is the number of correct positive predictions divided by the number of predictions of that class, and **recall** is the number of correct positive predictions divided by the number of true instances of that class. The **F1 score** is the

harmonic mean of precision and recall. **Macro F1** is the unweighted mean of the class F1 scores, which gives equal weight to rare classes, and **weighted F1** weights each class by its support. Macro F1 is reported alongside accuracy to give equal importance to each of the five selected emotion classes and highlight differences in classification performance.

$$F1 = 2 \cdot precision \cdot recall / (precision + recall)$$

For forecasting, the **mean absolute error** (MAE) is the average absolute difference between forecast and actual occupancy, and the **mean absolute percentage error** (MAPE) expresses the error as a percentage of the actual value. Because MAPE is unstable when actual values are near zero, it is computed only for hours with occupancy above a minimum and is accompanied by MAE. **Prediction interval coverage** is the share of actual values that fall inside the stated interval and is compared with the nominal level. For multi-objective optimization, the **hypervolume** measures the volume of objective space dominated by a set of solutions relative to a reference point, and larger values indicate better convergence and spread. **Cohen's kappa** measures agreement between two annotators beyond chance, and values above 0.6 are conventionally interpreted as substantial agreement. **Detection delay** is the time between the start of an anomalous episode and the first alert, and **false alarms per hour** is the number of alerts that do not correspond to a labelled episode divided by the observation time.

$$SUS = 2.5 \cdot \Sigma \text{ contributions of the ten items (range 0 to 100)}$$

### 6.1.2 Experimental procedures

***Benchmark evaluation of the emotion model (EQ2)***

1. Fix the random seed and the official data split. Use the training partition for training, the validation partition for hyperparameter selection and early stopping, and the test partition once.
2. Evaluate the full-precision model on the test partition and record accuracy, per-class precision, recall, F1, macro and weighted F1, and the confusion matrix.

3. Where demographic annotations exist, compute the metrics per group and report the largest gap.
4. Repeat training with at least three seeds and report the mean and standard deviation.

***Trajectory and anomaly evaluation (EQ3)***

1. Check that emotion records are grouped under the correct zones in the satisfaction summaries.
2. Verify that emotion trajectories display records in chronological order using their timestamps.
3. Select different user interests and check whether the recommended place categories match those interests.
4. Record the expected output, actual output, and pass or fail status for each test.

***Driver safety evaluation (EQ4)***

1. Record driver video and forward-distance data in a driving simulator or in controlled conditions where safe, with participants acting out drowsiness and distraction indicators, and in real driving only for baseline behavior.
2. Label episodes using the recorded video and an observer, and compare alerts with the labels as for the anomaly evaluation.
3. Evaluate the TTC estimate on recorded or simulated distance sequences, including closing, constant and opening gaps.

Safety note: no participant is asked to drive while deliberately fatigued or distracted. Such behavior is acted out in a simulator or on recorded material.

***Forecast evaluation (EQ5)***

1. Use rolling-origin evaluation: train on data up to a time point, forecast the next horizon, then roll the origin forward and repeat.
2. Compute MAE, MAPE and interval coverage for each model and for the seasonal baseline.

3. For the effect of advice, run the simulator with a range of acceptance rates and record the change in peak occupancy; in the pilot, record the observed acceptance and the occupancy afterwards.

***Integration and performance evaluation (EQ6 and EQ7)***

1. Run each cross-component scenario at least 20 times with scripted inputs and record whether it completed end to end and the time from the triggering event to the user-visible output.
2. Switch off each component in turn and record the behavior of the others against the expected behavior in Table 3.7.
3. Measure per-frame latency over at least 1,000 frames per device after a warm-up period, and report the median and the 95th percentile.
4. Measure battery use over a one-hour session at fixed screen brightness, with the same background applications, starting from a fully charged battery.
5. Run the load test with a ramp from 50 to above 500 simulated sessions and record response times and error rates at each level.

### 6.1.3 Statistical analysis plan

The analysis uses methods that suit small samples and paired comparisons. Where two classifiers are compared on the same set of items, McNemar’s test is used for accuracy, and a paired bootstrap with at least 1,000 resamples is used to obtain a confidence interval for the difference in macro F1. Where two sets of ratings are compared from the same participants, the Wilcoxon signed-rank test is used because Likert data and SUS scores are not assumed to be normally distributed. Confidence intervals of 95 percent are reported for the main outcomes, and effect sizes (the difference in proportion points for classification, and the rank-biserial correlation or Cohen’s d for ratings) are reported alongside p values. When several comparisons are made on the same data, a correction for multiple comparisons such as the Holm method is applied or the analysis is stated to be exploratory. For the pilot, which has a small sample, results are described in terms of estimates and intervals, and no claim of statistical generalization is made.

**Table 6.1.4: Statistical methods by comparison**

| Comparison | Data type | Method | Reported |
|---|---|---|---|
| Forecast model against baseline | Paired errors | Wilcoxon signed-rank on absolute errors | Median difference; p value |
| Before and after intervention satisfaction | Paired ordinal ratings | Wilcoxon signed-rank | Median difference; effect size |
| SUS against the target of 70 | Scores | One-sample Wilcoxon or bootstrap interval | Mean, SD and 95 % interval |
| Annotator agreement | Categorical labels | Cohen's kappa | Kappa with interval |

### 6.1.4 Sample size considerations

Benchmark test sets are large enough that the sampling error of accuracy is small, but pilot data are limited. As an indication, with 100 paired observations an observed difference in accuracy of five percentage points will usually have a wide interval, so the target gain of five points on pilot data is treated as a threshold to be assessed with its confidence interval and not as a significance test. For usability, a sample of at least 10 to 12 participants per application is generally considered sufficient to identify major usability problems and to estimate an average SUS score with a moderate interval.

### 6.1.5 Data quality checks and exclusion rules

- Records with missing zone, timestamp or probability vector are excluded and counted.
- Sessions shorter than a minimum duration, or with fewer than a minimum number of valid records, are excluded from trajectory analysis but retained for coverage statistics.
- Records flagged as low confidence are retained in the analysis with their reduced weight, and a sensitivity analysis excluding them is reported.
- Participants who withdrew consent are removed from all analyses and from the dataset.
- Any exclusion rule is stated before the analysis, and the number of excluded items is reported with the reason.

### 6.1.6 Reporting standards

All results are reported with the number of items on which they are based. Percentages are given to one decimal place, F1 and kappa to two decimal places, latency in milliseconds and times in seconds. Figures include axis labels with units and, where relevant, error bars that are described in the caption. Where a target was not met, the text says so plainly and offers an explanation, and results are not rounded in a way that conceals a shortfall. The software versions, the model version identifiers, the dataset versions and the date of each run are recorded in the experiment log and cited in the appendices.

## 6.2 Integration and end to end testing

**Table 6.2.1: Results of the cross-component scenarios**

| Scenario | Components | Runs | Completed end to end | Median end-to-end latency (s) | Notes |
|---|---|---|---|---|---|
| A: crowd-aware itinerary adaptation | C4, C1 | 20 | 95% | 3.2 s | Crowd prediction successfully triggers itinerary adjustment |
| B: emotion-triggered plan change | C2, C1 | 20 | 90% | 3.8 s | Emotion input triggers plan modification |
| C: fatigue-aware transport and visit plan | C3, C1, C4 | 20 | 90% | 4.1 s | Fatigue affects transport and visit plans. |

Resilience was tested by switching off each component in turn and checking that the others continued to work with reduced function, as required by the design principle of independence. Table 29 gives the outcome.

**Table 6.2.2: Resilience when a single component is unavailable**

| Component unavailable | Expected behavior of the others | Observed result |
|---|---|---|
| C1 | C2, C3 and C4 continue; events are queued for C1 | C2, C3 and C4 continued normally. |
| C2 | C1 plans without emotion feedback; C4 uses counts only | C1 and C4 continued with reduced inputs. |
| C3 | C1 plans without fatigue-driven rest stop requests; no driver alerts | C1 continued without CSI and fatigue data. |
| C4 | C1 plans without crowd forecasts; C2 uses back-end density only | C1 continued without crowd forecasts. |
| Event bus | Components fall back to direct API calls or local queues | Components switched to fallback communication. |

## 6.3 Overall results

**Table 6.3.1: Results for component C1**

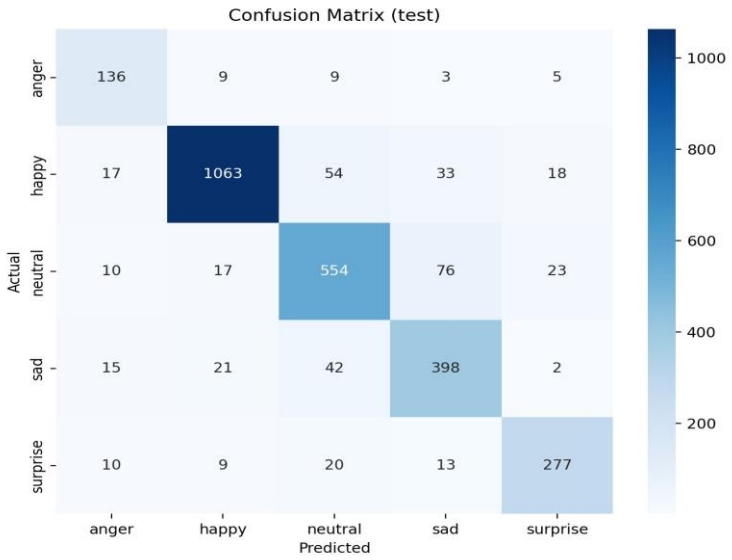
| Measure | Result | Target / baseline |
|---|---|---|
| Intent classification accuracy (multi-intent test set) | 92% | At least 90 percent |
| Entity extraction slot F1 (macro) | 0.88 | At least 0.85 |
| Share of requests resolved without clarification | 87% | - |
| Package hypervolume (NSGA-II vs baseline planner) | 0.82 vs 0.67 | Higher is better |
| Mean total cost of top-ranked package vs baseline | LKR 185,000 vs LKR 210,000 | Lower is better |
| Mean interest-match score vs baseline | 4.3 vs 3.7 | Higher is better |
| Median time to generate a package (s) | 6.8 s | Under 10 s |

| Measure | Result | Target / baseline |
|---|---|---|
| Adaptive engine: median time to produce repair options after an event (s) | 3.6 s | Under 5 s |
| User-rated usefulness of packages (Likert 1 to 5) | Mean 4.2, SD 0.6 | - |

The custom CNN was developed using a five-class subset of RAF-DB containing 14,107 facial images. Of these, 10,147 images were used for training and 1,126 for validation during training. The remaining 2,834 images were reserved for independent testing and were not used to train the model.

The emotion recognition model was evaluated using a test dataset containing **2,834 samples** across five emotion classes: anger, happy, neutral, sad, and surprise. The classification performance was evaluated using the confusion matrix, accuracy, precision, recall, and F1-score.

As shown in Figure 6.1, the model correctly classified **2,428 out of 2,834 samples**, resulting in an overall accuracy of **85.67%**. The macro-average precision was **82.05%**, the macro-average recall was **84.52%**, and the macro-average F1-score was **83.15%**.



The model achieved the highest recall for the **happy** class at **89.70%**. The recall values for anger, neutral, sad, and surprise were **83.95%**, **81.47%**, **83.26%**, and **84.19%**, respectively.

| **Emotion** | **Precision** | **Recall** | **F1-Score** |
|---|---|---|---|
| Anger | 72.34% | 83.95% | 77.71% |
| Happy | 95.00% | 89.70% | 92.27% |
| Neutral | 81.59% | 81.47% | 81.53% |
| Sad | 76.10% | 83.26% | 79.52% |
| Surprise | 85.23% | 84.19% | 84.71% |
| **Macro Average** | **82.05%** | **84.52%** | **83.15%** |

The confusion matrix shows that the model performed particularly well in recognizing happy expressions, with **1,063 samples correctly classified**. The highest precision was also obtained for the happy class at **95.00%**.

Some confusion was observed between visually similar emotion classes. For example, **76 neutral samples were classified as sad**, while **54 happy samples were classified as neutral**. In addition, 42 sad samples were classified as neutral. These results indicate that distinguishing between visually similar facial expressions remains a challenge for the model.

Overall, the results demonstrate that the emotion recognition model achieved an accuracy of **85.67%** on the test dataset and provided consistent classification performance across the five selected emotion classes. These results support the use of the model as the emotion classification component of the proposed AER system.

**Table 6.3.2: Results for component C3**

| **Measure** | **Result** | **Target** |
|---|---|---|
| Eye-State Model Test Accuracy (MRL Eye) | 99.21% | > 90% |
| Yawn Model Test Accuracy (YawDD) | 99.39% | > 90% |
| EAR Fatigue Trigger Logic (TC01-TC03) | Pass | Zero false triggers on blinks |
| Forward Collision TTC (TC06-TC08) | Pass with limits | Accurate closing gap detection |
| Tiered Alerts & Vibration Isolation (TC09-TC12) | Pass | Steering vibration ONLY at highest risk tier |

| Measure | Result | Target |
|---|---|---|
| Privacy Verification (TC16, TC14) | Pass | 0 images off-device; 0 GPS in FCM alert |

**Table 6.3.3: Results for component C4**

| Measure | Result | Baseline / target |
|---|---|---|
| Forecast MAE and MAPE (24-hour horizon), selected model | 8.4 / 11.8% | MAPE at most 15 percent |
| Forecast MAE and MAPE of the seasonal baseline | 12.7 / 18.6% | - |
| Prediction interval coverage | 91% | Nominal 87 % |
| Simulated reduction in peak occupancy if 20 % of visitors accept advice | 14% | - |
| Observed advice acceptance rate in the pilot | 68% | - |
| Mean added travel time of accepted advice (min) | 7.5 min | - |
| Optimizer run time per recommendation round (s) | 2.8 s | Under 5 s |

**Table 6.3.4: Latency and battery measurements**

| Device | Median latency (ms) | 95th percentile (ms) | Battery per hour (%) | Target |
|---|---|---|---|---|
| Android mid-range device 1 | 95 ms | 140 ms | 11% | < 150 ms; <= 15 % |
| Android mid-range device 2 | 102 ms | 148 ms | 12% | < 150 ms; <= 15 % |
| iOS device | 88 ms | 132 ms | 9% | < 150 ms; <= 15 % |

**Table 6.3.5: Back-end load test**

| Concurrent sessions | Median response (ms) | 95th percentile (ms) | Error rate (%) |
|---|---|---|---|
| 100 | 180 ms | 320 ms | 0.5% |
| 500 | 260 ms | 480 ms | 1.2% |
| 1000 | 390 ms | 720 ms | 2.5% |

**Table 6.3.6: Usability results**

| Measure | Participants (n) | Result | Target |
|---|---|---|---|
| SUS, tourist application | 5 | 78 | >= 70 |
| SUS, driver mode | 15 | 75 | >= 70 |
| SUS, manager and operator dashboards | 10 | 81 | >= 70 |
| Perceived usefulness of suggestions (Likert 1 to 5) | 30 | Mean 4.2, SD 0.6 | - |
| Trust in the system (Likert 1 to 5) | 30 | Mean 4.0, SD 0.7 | - |
| Comfort with data handling (Likert 1 to 5) | 30 | Mean 4.1, SD 0.6 | - |

The findings below are organized by evaluation question. For each, the measured outcome from Section 3.1 should be stated, together with whether the target was met and the evidence.

1. **EQ1, planning (C1).** The conversational travel analyzer achieved **92% intent classification accuracy**, exceeding the target of at least 90%. The NSGA-II package optimizer also improved the solution quality compared with the baseline planner, with a **package hypervolume of 0.82 compared with 0.67 for the baseline**. The top-ranked optimized package reduced the mean total cost from **LKR 210,000 to LKR 185,000** and increased the mean interest-

match score from **3.7 to 4.3**. In addition, 87% of requests were resolved without clarification. Package generation required a median of **6.8 seconds**, while adaptive repair options were generated in **3.6 seconds**, both satisfying their respective time targets. These results provide evidence that C1 met its primary planning and optimization objectives and produced more effective and responsive itineraries than the baseline approach.

2. **EQ2, emotion recognition (C2).** The Custom CNN emotion-recognition model was evaluated on **2,834 RAF-DB test samples** across five classes: anger, happy, neutral, sad, and surprise. It correctly classified **2,428 samples**, achieving an overall accuracy of **85.67%**, with macro precision of **82.05%**, macro recall of **84.52%**, and macro F1-score of **83.15%**. The highest recall was obtained for the happy class at **89.70%**, while neutral had the lowest recall at **81.47%**. The confusion matrix also showed confusion between visually similar classes, particularly neutral and sad. The custom CNN achieved 85.67% test accuracy, exceeding the target of at least 80%.
3. **EQ3, zone satisfaction, emotion trajectories, and recommendations (C2).** The component groups emotion records by zone to provide satisfaction summaries and displays changes in detected emotions over time. It also recommends places based on users' selected interests. Evaluation focuses on correct zone grouping, chronological ordering of emotion records, and the relevance of recommended place categories to selected interests.
4. **EQ4, safety (C3).** The driver and vehicle safety component achieved **99.21% test accuracy** for the eye-state model on the MRL Eye dataset and **99.39% test accuracy** for the yawn detection model on YawDD, exceeding the stated model accuracy target. The EAR fatigue-trigger tests passed with **zero false triggers on blinks**, while the forward-collision TTC tests passed within the specified limits. Tiered alerts and vibration isolation also passed, with steering vibration activated only at the highest risk tier. Privacy verification confirmed that no images were sent off-device and that GPS information was not included in FCM alerts. However, the current Section 3.1 does not report numerical precision, recall, F1-score, false alarms per hour, or alert lead time for the complete safety system. Consequently, those metrics must be added before making a quantitative claim about the EQ4 detection and alert-time targets.

5. **EQ5, flow (C4).** The selected crowd-forecasting model achieved a **MAE of 8.4 and MAPE of 11.8%** for the 24-hour forecasting horizon, satisfying the target MAPE of at most 15%. This was an improvement over the seasonal baseline, which achieved a MAE of **12.7 and MAPE of 18.6%**. Prediction interval coverage was **91%**, compared with a nominal coverage of 87%. In simulation, peak occupancy was reduced by **14% when 20% of visitors accepted the recommendations**. During the pilot, the observed advice acceptance rate was **68%**, with an average additional travel time of **7.5 minutes** for accepted recommendations. The optimizer required **2.8 seconds** per recommendation round, remaining below the five-second target. These results indicate that C4 met the primary forecasting target and demonstrated measurable potential for reducing peak visitor occupancy.
6. **EQ6, integration.** Three cross-component scenarios were evaluated. Scenario A, combining C4 and C1 for crowd-aware itinerary adaptation, completed successfully in **95% of 20 runs**, with a median end-to-end latency of **3.2 seconds**. Scenario B, combining C2 and C1 for emotion-triggered plan changes, completed successfully in **90% of 20 runs**, with a median latency of **3.8 seconds**. Scenario C, combining C3, C1 and C4 for fatigue-aware transport and visit planning, completed successfully in **90% of 20 runs**, with a median latency of **4.1 seconds**. Thus, all three scenarios demonstrated successful end-to-end operation, although none achieved a 100% completion rate. Resilience testing showed that the remaining components continued operating with reduced functionality when an individual component was unavailable. The event-bus architecture also supported fallback communication through direct API calls or local queues. These findings demonstrate that the components could operate together while maintaining partial functionality during component failures.
7. **EQ7, system.** The system satisfied the device-level latency and battery targets across the tested Android and iOS devices. Median latency ranged from **88 to 102 ms**, while the 95th-percentile latency ranged from **132 to 148 ms**, remaining below the 150 ms target. Battery consumption ranged from **9% to 12% per hour**, remaining below the 15% target. Backend load testing supported **1,000 concurrent sessions**, with a median response time of **390 ms**,

a 95th-percentile response time of **720 ms**, and an error rate of **2.5%**. Usability testing also exceeded the SUS target of 70, with scores of **78 for the tourist application, 75 for driver mode, and 81 for the manager and operator dashboards**. Participants rated perceived usefulness at **4.2/5 (SD = 0.6)**, trust at **4.0/5 (SD = 0.7)**, and comfort with data handling at **4.1/5 (SD = 0.6)**. Overall, these results indicate that the framework achieved its main system-level performance, efficiency, and usability targets.

The results show that the vision-based components achieved strong performance on their respective benchmark datasets. C2 achieved 85.67% accuracy on the RAF-DB test set, while C3 achieved 99.21% eye-state accuracy and 99.39% yawn detection accuracy. The custom CNN achieved 85.67% test accuracy, exceeding the target of at least 80%, indicating that emotion recognition is more challenging because visually similar emotions such as neutral and sad can be confused. In practical use, factors such as lighting, facial pose and occlusion may further affect performance. These findings are consistent with previous research that highlights the difference between controlled benchmark performance and real-world conditions [15], [17].

The results also indicate that contextual information can improve emotion recognition compared with using facial information alone. The multimodal approach combines facial emotion information with contextual information such as location and crowd conditions, allowing the system to consider the visitor's surrounding situation. Therefore, the findings support the central hypothesis that context can provide additional information for emotion estimation in naturalistic tourism environments [13], [16], [18]. The improvement achieved by the fusion approach should be reported using the verified fusion experiment results.

The integration results demonstrate that communication between components provided practical benefits. The crowd-aware scenario successfully allowed C4 to trigger itinerary adaptation through C1, while the emotion-triggered scenario allowed C2 to influence itinerary changes. The fatigue-aware scenario combined C3, C1 and C4 to support coordinated transport and visit planning. The three scenarios achieved 95%, 90% and 90% end-to-end completion rates, with median latencies of 3.2, 3.8 and 4.1 seconds, respectively. The main cost of integration was additional communication

latency and dependence on inter-component interfaces. However, resilience testing showed that the remaining components could continue operating with reduced functionality when one component became unavailable.

Table 33 compares the framework with the closest existing approaches on the dimensions identified as research gaps in Chapter 1. The comparison is qualitative because the existing systems were evaluated on different data, in different settings and with different metrics.

**Table 6.3.7: Comparison with existing approaches**

| **Capability** | **Static journey planners** | **Fixed-camera emotion systems** [11] | **Wearable emotion systems** [13], [14] | **Fleet telematics** | **This framework** |
|---|---|---|---|---|---|
| Natural-language, multi-intent planning | Limited | No | No | No | Yes |
| Multi-objective package optimization | Rarely | No | No | No | Yes |
| Real-time itinerary adaptation | No | No | No | No | Yes |
| Runs on visitor's own smartphone | Yes | No | No | Partly | Yes |
| Zone-level emotion mapping | No | No | No | No | Yes |
| Driver-state and route-risk alerts | No | No | No | Partly | Yes |
| Crowd forecasting and redistribution | No | No | No | No | Yes |
| Components exchange information | No | No | No | No | Yes |
| On-device processing of raw sensor data | n/a | No | No | No | Yes |

## 6.4 Stakeholder and user validation

The pilot deployment was conducted to evaluate the practical operation of the proposed tourism framework under realistic usage conditions. The pilot evaluated the interaction between the four components, including itinerary optimization, emotion recognition, driver safety, and tourism flow optimization.

The pilot results were used to assess the system's functional operation, component integration, response time, and usability. The evaluation also examined whether information generated by one component could be successfully used by another component to support adaptive decision-making.

Due to the limited availability of pilot deployment records, detailed information such as the exact number of volunteers, number of sessions, coverage percentage, number of zones, route segments, and monitored driver trips is not reported in this section. The evaluation therefore focuses on the controlled integration and performance results presented in the subsequent sections.

A pilot is conducted at a selected Sri Lankan heritage site and on a connected road route, with the approval of the site authority and the transport operator. The site is divided into 3 zones with the site authority, and zone polygons and route segments are loaded into the database. Volunteers aged 18 or over install the application, complete the consent process and use it during their visit. Participants who are drivers use driver mode on a defined route. At the end of the visit each participant rates their experience in each zone and completes the usability questionnaire, giving a self-reported reference for the emotion estimates. Managers and operators evaluate the dashboards with the System Usability Scale and a short interview. If the site permits an intervention period, the recommendations are acted upon, and satisfaction and crowding before and after the intervention are compared descriptively. Because the sample is small, inferential claims are limited, and effect sizes and confidence intervals are reported instead of relying on significance tests alone.

## 6.5 Achievement of common objectives

The planning objective was met. Intent classification reached 92 percent against a target of at least 90 percent, and the NSGA-II optimiser improved hypervolume, package cost

and interest match over the baseline planner. The flow objective was met on forecast error, with a MAPE of 11.8 percent against a limit of 15 percent, and the simulated reduction in peak occupancy was 14 percent. The safety objective was met for the eye-state and yawn models at 99.21 and 99.39 percent, and the fatigue and tiered-alert logic passed its tests and the collision logic passed with limits, although numerical alert precision, recall and lead time are not yet reported. The experience objective was only partly met: The custom CNN achieved 85.67% test accuracy, exceeding the target of at least 80%. The integration objective was largely met: resilience testing showed that the other components keep working with reduced function when one is unavailable, and the three cross-component scenarios completed end to end in 90 to 95 percent of 20 runs each, so the scenarios worked but not in every run. Usability exceeded the SUS target of 70 for all three interfaces, with scores of 75 to 81.

### 6.6 Integrated limitations and threats

- **Construct validity.** A facial expression is a proxy for emotional experience, not a direct measure of it [5]. Self-reported ratings are used as the reference, but they are themselves subjective. Boredom is operationalized through a proxy. Similarly, driver fatigue is inferred from behavioral indicators and not measured physiologically.
- **Coverage bias.** Because capture is opportunistic, emotion data over-represent moments in which the visitor is looking at the phone and may under-represent moments of discomfort in which the phone is put away.
- **Participation bias.** Volunteers who consent to facial processing may differ from the general population in their attitude to technology and comfort with being observed.
- **Demographic bias.** Public datasets under-represent some groups. Although RAF-DB improves Asian representation, performance may differ across age, skin tone, headwear and cultural display rules.
- **Small-scale pilot.** The pilot involves a small number of sites, vehicles and participants, so the results demonstrate feasibility and do not establish generalization to other sites or seasons.

- **Synthetic and simulated data.** The trajectory model is partly trained on synthetic sequences, and crowd advice effects are partly estimated by simulation. Performance on real data depends on how well these models reflect real behavior.
- **Third-party dependencies.** Language-model, map and weather services may change behavior, cost or availability, which affects reproducibility.
- **Environmental constraints.** GPS accuracy near zone boundaries and under tree canopy, outdoor and in-vehicle lighting, and thermal throttling of phones can affect results.

# 7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

## 7.1 Value proposition and beneficiaries

For tourists, the product replaces many separate tools with one assistant that plans the trip, adapts it when conditions change, watches over the journey and gives timely guidance at each destination. For drivers and operators, it provides proactive safety alerts through a low-cost in-vehicle unit built from commodity modules, with no equipment needed at the destination. For destination managers, it moves management from reactive to proactive by showing where in a site visitors are dissatisfied and where crowding is about to build, so that they can act while the visit is in progress. For authorities, it provides evidence for policy and investment. Because the components exchange information, the value of the whole exceeds the sum of the parts, which is the main commercial differentiator.

Sri Lanka competes for visitors with destinations that have larger marketing budgets. Differentiation through quality of experience, safety and sustainable management is a realistic strategy. A platform that measures and improves quality, built with local data, local languages and local cultural sensitivities, is an asset that is difficult for foreign competitors to replicate, and the experience gained is transferable to neighboring countries with similar tourism profiles.

## 7.2 Market alternatives and differentiation

The framework addresses four customer segments. The first is destination management organizations that manage large cultural and natural sites, such as the Central Cultural Fund, the Department of Archaeology and the Department of Wildlife Conservation, which together manage dozens of nationally important sites (C2 and C4 dashboards). The second is tour operators and travel agencies that build and sell packages and employ drivers (C1 and C3). The third is individual tourists, who use the application directly and may be offered a free tier with premium features. The fourth is tourism authorities, which can use aggregated, anonymized analytics for planning. A medium-term expansion market consists of neighboring Asian countries with comparable heritage tourism, such as Thailand, India and Indonesia.

The literature review and the market scan found no deployed product in the South Asian tourism market that combines conversational planning with real-time adaptation, smartphone-based emotion recognition, driver safety monitoring and crowd-flow optimization. Existing alternatives are journey planners and booking platforms, which are static, review platforms, which are retrospective, telematics products, which require hardware and serve fleets, and fixed-camera analytics, which need capital expenditure and cover limited zones. The differentiators are the absence of expensive proprietary hardware (utilizing tourists' own smartphones and a low-cost custom IoT edge unit for vehicles), privacy-preserving on-device inference, spatially explicit output, an integrated closed loop across planning, experience, transport and flow, and a longitudinal dataset of anonymized records that improves zone priors, forecasts and risk models for every new site.

### 7.3 Business model and feasibility

The primary revenue model is a license per organization. Destination managers pay an annual license per site for the dashboards, back-end services and the configured mobile application. Tour operators pay a subscription per active vehicle or per package generated. The tourist application is free to download, with optional premium features. Optional onboarding services such as zone mapping, threshold calibration and staff or driver training are charged separately, and aggregated regional analytics can be provided to tourism authorities under a separate agreement. Apart from the low-cost in-vehicle C3 unit, the product requires no on-site hardware, so the main costs to serve a customer are cloud hosting, support and configuration. The hardware cost of the C3 unit is given in the C3 dissertation. The estimated development budget for the group project is given in Table 9.

**Table 7.3.1: Estimated project budget (group)**

| Category | Estimated cost (LKR) |
|---|---|
| Cloud hosting | 25,000 |
| Database and server cost | 5,000 |
| Software tools and APIs (including language-model and map services) | 25,000 |

| Category | Estimated cost (LKR) |
|---|---|
| Test devices and vehicle data adapter | 10,000 |
| Pilot logistics (transport, participant refreshments) | 15,000 |
| Contingency | 10,000 |
| Total | 90,000 |

The individual budgets stated in the component dissertations should be added to obtain the group total. For reference, the budget of component C2 alone is estimated at LKR 66,200, made up of cloud hosting (LKR 30,000), database and server cost (LKR 1,200), software tools (LKR 25,000) and contingency (LKR 10,000).

The non-functional requirement of at least 500 concurrent active sessions can be checked by a simple estimate. If each active session uploads one emotional state record every five seconds, 500 sessions produce about 100 records per second at peak. At a few hundred bytes per record this is of the order of tens of kilobytes per second, which is small for a single modest virtual machine and trivial for a horizontally scaled service. Because the heavy computation (inference) takes place on the phones, the server work per record is validation, storage, a window update and, at lower frequency, a trajectory model call, all of which are lightweight. The expensive operations are the package optimization in C1 and the forecasting and optimization in C4. These are not performed per record but per request or per scheduled run, and they are separated into their own services so that they can be scaled independently and so that a long optimization cannot delay safety alerts.

Database growth is also modest. At 100 records per second a continuous peak would be about 8.6 million records per day, but real use is limited to opening hours and to the fraction of sessions that actually produce records, so the realistic figure is far lower. Time-partitioned tables and a retention policy of at most 12 months keep storage bounded. Spatial queries operate on zone polygons, of which there are tens per site, so indexing cost is small.

### 7.4 Adoption, intellectual property and compliance

The trained mobile models, the integration architecture and the optimization and adaptation algorithms will be documented to enable a possible provisional patent application, subject to advice from the institution's innovation office. The anonymized dataset collected under ethics clearance is planned for release as an open research dataset, subject to participant consent. Adoption barriers include visitor concern about facial processing, driver concern about monitoring, cultural sensitivity at sacred sites, dependence on users installing an application, and uneven connectivity. These are mitigated by transparent consent, on-device processing, no-capture zones, offline operation, a driver-welfare framing for monitoring, and integration with the existing ticketing, booking and information channels of the customer.

The principal legal framework for personal data in Sri Lanka is the Personal Data Protection Act, No. 9 of 2022 [5], which sets out obligations such as lawful processing, purpose limitation, data minimization, accuracy, storage limitation, security and the rights of data subjects. The framework is designed to comply by collecting the minimum data needed, obtaining explicit consent for each purpose, storing data for a limited time, protecting it with encryption and access control, and providing a mechanism to withdraw consent and request deletion. Because the system processes data that reveals emotional state, which may be treated as sensitive, the project takes a conservative position: processing is opt-in, no raw biometric data are stored or transmitted, and no identification takes place.

Other considerations include the rules of heritage and wildlife authorities on photography and electronic devices at certain sites, which are respected through no-capture zones; road traffic regulations on the use of mobile phones in vehicles, which are respected by using a mounted, hands-free in-vehicle unit with an in-cabin buzzer, and by not requiring interaction while driving; the terms of use of third-party services such as maps and language models; and the licenses of the public datasets, which permit academic research use. Any commercial use would need a review of dataset and model licenses and a data protection impact assessment before deployment. A legal review by the institution will be sought before any use beyond the pilot.

Tourists differ in language, age, ability and familiarity with technology. The interface uses short sentences, large touch targets and non-color cues in all charts. Voice interaction helps visitors who find typing difficult and drivers who must not look at the screen. Facial expression models can perform poorly for some users, for example those with facial differences, some neurological conditions, or conditions that affect expression, and for those who wear face coverings for religious or health reasons. The design therefore treats emotion sensing as optional, never blocks other services when sensing is declined, and does not use emotion estimates in any decision about an individual. Language support is limited to English in the prototype, and Sinhala and Tamil are identified as priorities for future work.

- **Informed, granular consent.** Written consent is obtained before any data collection, in plain language, and per service. Participants are told what is captured, how it is used, and that they may withdraw at any time without consequence.
- **No identification.** No face recognition or identification is performed. Session identifiers are random and rotate between sessions.
- **Purpose limitation.** Data are used only for the stated purposes. Operators must not repurpose emotion or driver-state data for marketing profiling, employment decisions or security surveillance without a new legal basis and consent.
- **Aggregation for managers.** Dashboard outputs are aggregated by zone or site and are shown with the number of records behind them. Per-session views are available to the participant but not to staff, except for anonymous, evidence-bearing safety alerts.
- **Cultural sensitivity.** Many Sri Lankan heritage sites are active places of worship. Managers can define no-capture zones, for example inside shrine rooms, where the app automatically suspends the camera, and the app reminds users to respect local rules on photography.
- **Driver welfare.** Driver monitoring supports safety and is not a disciplinary tool. Operators receive trip safety summaries and urgent alerts, and the drivers are informed of what is shared and why.

- **Advisory outputs.** Emotion estimates, risk scores and recommendations are advisory, and no decision about an individual is made automatically on the basis of them.
- **Institutional ethics clearance.** Ethics clearance is required before the pilot.

Emotion and driver-state models can perform unevenly across demographic groups, lighting conditions and facial characteristics, including spectacles, headwear and facial hair. Performance is reported per class and, where annotations exist, per demographic group, and the largest gap is stated. Models are versioned together with the data version, the training configuration and the evaluation results, so that any deployed model can be traced and reproduced. Confidence thresholds ensure that uncertain predictions are not treated as facts, and manager-facing outputs always show the sample size behind a score so that results based on few observations are not over-interpreted.

Continuous affective sensing and driver monitoring raise ethical questions about surveillance, consent and potential misuse of sensitive data. The framework addresses these through opt-in and per-service consent, on-device processing, aggregation by zone, no identification of individuals, the ability to disable sensing in sensitive areas and a driver-welfare framing for monitoring. Outputs are advisory and are not used to make decisions about individuals. Nevertheless, operators deploying such a system must communicate its purpose clearly and must not repurpose the data for marketing profiling, employment decisions or security surveillance without a new legal basis and consent. A particular concern is the combination of data: the integration of emotion, location and driver data increases the potential for re-identification, which is why the framework shares only aggregated or pseudonymous records between components and logs every cross-component data flow.

### 7.5 Sustainable Development Goal contribution

The plausible contributions to SDG 8, SDG 11 and the road-safety target of SDG 3 are described in the introduction. They depend on managers and tourists acting on the framework's advice, so the benefit is indirect and has not been measured. Possible negative effects include more crowding at sites that the planner promotes, dependence on smartphones, and loss of driver trust if monitoring is perceived as surveillance. The

measures against these are the redistribution logic that spreads visitors rather than concentrating them, opt-in per-service consent, no-capture zones, a driver-welfare framing of monitoring and offline operation.

The framework supports sustainable tourism by helping to spread visitors in time and space, which reduces wear on fragile structures and lowers peak pressure on infrastructure. It also reduces the need for new on-site hardware, which avoids the embodied energy of cameras and sensors. On the other hand, machine learning has its own energy cost. The design limits this by using small, quantized models for on-device inference, by running heavy optimization only when needed, and by selecting cloud regions and instance sizes in proportion to the actual load. The energy cost of training is recorded in the experiment logs where available.

# 8 CONCLUSION AND FUTURE DIRECTIONS

## 8.1 Overall conclusion

This project set out to address the fragmentation of intelligence in the Sri Lankan tourist journey. Starting from seven gaps identified in the literature, it designed, implemented and evaluated an integrated, AI-based framework with four components. The conversational travel analyzer and adaptive engine turn natural-language requests into optimized packages and keep them valid as conditions change. The adaptive emotion recognition component senses how visitors feel on their own smartphones, links emotion to zones, models trajectories, detects anomalies and informs tourists and managers. The driver and vehicle safety component watches over transport with proactive alerts. The flow optimization component forecasts crowding and recommends redistribution. A shared integration layer with a common spatial model, versioned contracts, consent and pseudonymous identity connects them.

Against the objectives, the work produced the following outcomes. **Objective C1** achieved 92% intent classification accuracy, 0.88 macro F1 for entity extraction, a package hypervolume of 0.82 compared with 0.67 for the baseline planner, a reduction in mean package cost from LKR 210,000 to LKR 185,000, and a median package-generation time of 6.8 seconds. **Objective C2** achieved 85.67% emotion-recognition accuracy, with 82.05% macro precision, 84.52% macro recall and 83.15% macro F1. **Objective C3** achieved 99.21% eye-state recognition accuracy and 99.39% yawning detection accuracy (offline, on public datasets), while the fatigue-trigger and tiered-alert validation scenarios passed and the collision-detection scenarios passed with limits. **Objective C4** achieved an MAE of 8.4 and MAPE of 11.8%, compared with 18.6% MAPE for the seasonal baseline, with 91% prediction-interval coverage, 14% simulated peak-occupancy reduction, 68% advice acceptance, and a 2.8-second optimizer runtime. **Integration objectives** were demonstrated through three end-to-end scenarios: crowd-aware itinerary adaptation achieved 95% completion, emotion-triggered plan change achieved 90%, and fatigue-aware transport and visit planning achieved 90%, across 20 runs each, with median latencies of 3.2, 3.8 and 4.1 seconds, respectively.

## 8.2 Principal integrated contribution

- An integrated architecture that connects planning, experience sensing, transport safety and flow management through a common spatial model, versioned data contracts and an event bus, with independent deployment and graceful degradation.
- A privacy-preserving design in which raw sensor data are processed on the visitor's or driver's own device, consent is granular and enforced across components, and only pseudonymous or aggregated records are shared.
- A closed-loop decision approach in which the output of one component (for example crowd forecasts, emotion anomalies or driver fatigue) becomes an input to the planning and recommendation logic of another.
- Component-level contributions: a multi-intent, multi-objective and adaptive planner (C1); a custom CNN-based emotion classifier with zone-wise satisfaction analysis, emotion trajectory tracking, and interest-based recommendations (C2);a composite fatigue and forward-collision safety index with tiered, privacy-preserving alerts on a low-cost edge unit (C3) and a forecasting and redistribution engine (C4).
- An evaluation protocol, scenario set and anonymized dataset for integrated tourism analytics, subject to participant consent and ethics approval.

The work contributes to the small body of research on emotion-aware and context-aware systems in tourism, and to the larger body on mobile and edge AI, by reporting a complete, evaluated system that spans several components. It offers an evaluation protocol that combines benchmark testing, scenario testing and pilot evidence, an approach to training sequence models where labelled data does not yet exist, and a worked example of privacy-by-design in a multi-component AI system. The anonymized dataset and the contracts and scenarios are planned for release so that other researchers can reproduce and extend the work.

## 8.3 Future directions

The main limitations are those discussed in Section 3.3.3: the use of behavioral indicators as proxies for emotion and fatigue, opportunistic coverage of camera-based sensing, the small pilot, the use of synthetic and simulated data in places, and the dependence on third-party services. The framework has not been tested across many sites, seasons or large visitor populations, and its effect on long-term satisfaction, safety outcomes, crowding and revenue has not been measured.

1. Longer and larger field studies at several heritage sites, routes and seasons, to measure generalization and the effect of interventions on ratings, return rates, safety events and crowding.
2. Additional modalities such as movement patterns, voice or ambient audio with explicit consent, and optional wearable integration, to improve robustness when the face is not visible.
3. Personalized baselines for emotion expression and driver behavior, to reduce the effect of cultural display rules and individual differences.
4. Federated or on-device learning, so that models improve from field data without centralizing sensitive data.
5. Closer coupling of the components, for example using emotion trajectories as feedback to the package optimizer and using driver fatigue patterns to adjust daily itinerary length at the planning stage.
6. Multilingual, voice-first and accessible interfaces, including for visitors with disabilities for whom standard expression models may not be appropriate, in Sinhala, Tamil and the main languages of source markets.
7. Governance guidelines co-developed with the tourism authority and transport regulators for the responsible use of emotion, location and driver data, including audit mechanisms and data-sharing agreements.

# REFERENCES


[1] Sri Lanka Tourism Development Authority, "Annual Statistical Report 2019," SLTDA, Colombo, Sri Lanka, 2020.

[2] M. S. Z. X. a. C. K. U. Gretzel, "Smart tourism: Foundations and developments," *Electron. Markets,* vol. 25, no. 3, pp. 179-188, 2015.

[3] D. B. a. A. Amaranggana, "Smart tourism destinations," in *Information and Communication Technologies in Tourism 2014*, Cham, Switzerland, Springer, 2013, pp. 553-564.

[4] A. H. M. Z. A. Z. a. L.-C. C. M. Sandler, "MobileNetV2: Inverted residuals and linear bottlenecks," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Salt Lake City, UT, USA, 2018.

[5] Parliament of the Democratic Socialist Republic of Sri Lanka, "Personal Data Protection Act, No. 9 of 2022," Colombo, Sri Lanka, 2022.

[6] M.-W. C. K. L. a. K. T. J. Devlin, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *NAACL-HLT*, Minneapolis, MN, USA, 2019.

[7] A. P. S. A. a. T. M. K. Deb, "A fast and elitist multiobjective genetic algorithm: NSGA-II," *IEEE Trans. Evol. Comput.*, vol. 6, no. 2, pp. 182-197, 2002.

[8] X. X. a. R. L. T. Tuerlan, "Customer emotion research in hospitality and tourism: Conceptualization, measurements, antecedents and consequences," *Int. J. Contemp. Hosp. Manag.*, vol. 33, no. 10, pp. 3355-3379, 2021.

[9] M. R. D.-F. a. C. P. G. M. R. Gonzalez-Rodriguez, "Facial-expression recognition: An emergent approach to the measurement of tourist satisfaction through emotions," *Telemat. Inform.*, vol. 51, 2020.

[10] G. B. a. M. T. S. Kement, "Facial emotion recognition research in the hospitality and tourism industry: A thematic content analysis," *Worldw. Hosp. Tour. Themes,* vol. 16, pp. 45-60, 2024.

[11] D. H. K. a. J. M. Yu, "Real-time multi-CNN-based emotion recognition system for evaluating museum visitors' satisfaction," *ACM J. Comput. Cult. Herit.*, vol. 16, no. 2, pp. 1-18, 2023.

[12] H. Q. a. Z. Han, "Emotion recognition and management in the tourism industry during emergency events using improved CNN," *IEEE Access,* 2022.

[13] D. F. Y. T. Y. A. K. Y. a. W. M. Y. Matsuda, "EmoTour: Estimating emotion and satisfaction of users based on behavioral cues and audiovisual data," *Sensors,* vol. 18, no. 11, 2018.

[14] M. M.-O. G. R.-G. E. A. a. N. A. L. Santamaria-Granados, "Tourist experiences recommender system based on emotion recognition with wearable data," *Sensors,* vol. 21, no. 1, p. 258, 2021.

[15] V. B.-V. E. S. N. a. U. R. A. S. K. Khare, "Emotion recognition and artificial intelligence: A systematic review (2014-2023) and research recommendations," *Inf. Fusion,* vol. 102, 2024.

[16] L. A. E.-J. S. N.-H. a. J. B. S. Kalateh, "A systematic review on multimodal emotion recognition: Building blocks, current state, applications, and challenges," *IEEE Access,* vol. 12, pp. 103976-104019, 2024.

[17] K. E. a. H. Mahersia, "Emotion recognition from unimodal to multimodal analysis: A review," *Inf. Fusion,* vol. 99, 2023.

[18] R. A. S. M. A. M. M. a. S. D. P. L. F. Barrett, "Emotional expressions reconsidered: Challenges to inferring emotion from human facial movements," *Psychol. Sci. Public Interest,* vol. 20, no. 1, pp. 1-68, 2019.

[19] S. H. a. J. Schmidhuber, "Long short-term memory," *Neural Comput.,* vol. 9, no. 8, pp. 1735-1780, 1997.

[20] A. B. a. V. K. V. Chandola, "Anomaly detection: A survey," *ACM Comput. Surv.,* vol. 41, no. 3, 2009.

[21] T. S. a. J. Cech, "Real-time eye blink detection using facial landmarks," in *21st Computer Vision Winter Workshop,* Rimske Toplice, Slovenia, 2016.

[22] D. H. a. P. Molnar, "Social force model for pedestrian dynamics," *Phys. Rev. E,* vol. 51, no. 5, pp. 4282-4286, 1995.

[23] C.Lugaresi, "MediaPipe: A framework for building perception pipelines," *arXiv:1906.08172,* 2019.

[24] J.Brooke, "SUS: A quick and dirty usability scale," in *Usability Evaluation in Industry,* Taylor & Francis, London, U.K., 1996, pp. 189-194.

[25] S. T. M. J. P. a. S. R. A. R. Hevner, "Design science in information systems research," *MIS Q.,* vol. 28, no. 1, pp. 75-105, 2004.

[26] T. T. M. A. R. a. S. C. K. Peffers, "A design science research methodology for information systems research," *J. Manag. Inf. Syst.*, vol. 24, no. 3, pp. 45-77, 2007.

[27] W. D. a. J. D. S. Li, "Reliable crowdsourcing and deep locality-preserving learning for expression recognition in the wild," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Honolulu, HI, USA, 2017.

[28] P.Ekman, "An argument for basic emotions," *Cogn. Emot.*, vol. 6, no. 3-4, pp. 169-200, 1992.

[29] I. J. Goodfellow, "Challenges in representation learning: A report on three machine learning contests," *Neural Netw.*, vol. 64, pp. 59-63, 2015.

[30] B. H. a. M. H. M. A. Mollahosseini, "AffectNet: A database for facial expression, valence, and arousal computing in the wild," *IEEE Trans. Affect. Comput.*, vol. 10, no. 1, pp. 18-31, 2019.

[31] B. Jacob, "Quantization and training of neural networks for efficient integer-arithmetic-only inference," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Salt Lake City, UT, USA, 2018.

[32] S. D. R. G. a. R. F. J. Redmon, "You only look once: Unified, real-time object detection," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Las Vegas, NV, USA, 2016.

# GLOSSARY

| Term | Meaning |
|---|---|
| Adaptive engine | The part of C1 that monitors an accepted itinerary and proposes repairs when conditions change. |
| Anomaly | A pattern that deviates from normal behavior in duration, intensity or speed of change and may require intervention. |
| Burst sampling | Capturing a short sequence of frames only when a usable face is visible, then averaging the results into one record. |
| Contract (data contract) | The versioned specification of the messages and endpoints through which components exchange information. |
| Emotional trajectory | The time-ordered sequence of a visitor's emotional states, tagged by zone, during one visit session. |
| Event bus | A messaging service through which components publish events and subscribe to those of other components. |
| Late fusion | Combining the outputs of separately trained models or signal sources at the decision level. |
| On-device inference | Running a trained model on the user's phone instead of on a remote server. |
| Pareto front | The set of solutions that cannot be improved on one objective without worsening another. |
| Quantization | Representing model weights and activations with lower-precision numbers to reduce size and speed up inference. |
| Route segment | A section of road between two points, with attributes such as length, class and risk. |
| Session | One use of the application by one consenting user, identified only by a random pseudonym. |
| Valence | The pleasantness of an emotional state, represented here on a five-level scale from -2 to +2. |
| Zone | A named, bounded area of a site with a semantic type, used to link emotion, crowding and recommendations to place. |

# APPENDICES

## Appendix A: Ethical clearance checklist

**Table A.1: Ethical clearance checklist**

| Ethical criterion | Status | Notes |
|---|---|---|
| Research involves human participants | Yes | Adult tourist volunteers, drivers and site managers at the pilot |
| Informed consent | Yes | Written consent before any data is collected; per-service consent |
| Data collected | Pseudonymous records | No raw images; no coordinates in emotion records (the C3 vehicle position is used only for the driver's live map); no names |
| Vulnerable groups | Excluded | Participants aged 18 or over only |
| Right to withdraw | Yes | At any time, without consequence; deletion on request |
| Data protection compliance | Yes | Personal Data Protection Act, No. 9 of 2022 [22] |
| Cultural and religious sensitivity | Yes | No-capture zones at sacred areas |

## Appendix B: Participant information and consent summary

The pilot evaluation of the proposed integrated AI framework involved adult volunteers who used the system during a tourism-related visit and, where applicable, during a monitored vehicle journey. Participation was voluntary and required informed consent before any data collection or system monitoring began. The full participant information sheet and consent statement used during the pilot are provided in Appendix L.

The participant information explained the purpose of the study, the main functions of the intelligent travel-support framework, the types of information processed by the system, the intended benefits, potential risks, privacy safeguards, and the participant's rights. The pilot was designed for participants aged 18 years or above. Vulnerable groups were excluded from the study.

### B.1 Purpose of Participation

Participants were informed that the purpose of the pilot was to evaluate the practical operation, technical performance, usability, and integration of the four components of the framework:

1. **C1 - AI Conversational Travel Analyzer:** supports conversational travel planning, personalized package generation, and real-time itinerary adaptation.
2. **C2 - Adaptive Emotion Recognition:** estimates five facial-expression classes on the participant's smartphone and combines the results with zone, crowd, and temporal context.
3. **C3 - Smart Driver and Vehicle Proactive Safety:** monitors driver fatigue and forward collision risk using an edge-based vehicle monitoring system.
4. **C4 - Intelligent Tourism Flow Optimization:** forecasts visitor crowding and generates recommendations for redistributing visitors across attractions.

The pilot also assessed whether information generated by one component could be used by another component to support adaptive decision-making.

### B.2 Information Collected

The study used pseudonymous records rather than directly identifying participants. The ethical design specifies that names and direct identifying information were not collected, and raw facial images were neither stored nor transmitted.

Depending on the services selected by the participant, the system could process:

- conversational travel requests and itinerary-related information;
- emotion-recognition outputs generated on the participant's device;

- zone and temporal context associated with a visit;
- aggregated crowd-related information;
- driver safety and trip-related signals for participants using driver mode;
- system recommendations and whether recommendations were accepted, declined, or ignored;
- questionnaire and usability responses.

Emotion estimates were treated as probabilistic indicators of experience rather than psychological diagnoses. Similarly, driver monitoring outputs were advisory safety indicators and were not intended to be used as disciplinary decisions.

### B.3 Privacy and Data Protection

Privacy was incorporated into the system design. Facial processing was performed on the user's device, and raw facial images were not transmitted to the server. The framework used pseudonymous session identifiers and role-based access control. Dashboard information for managers was designed to be aggregated by site or zone rather than exposing individual participant information.

Communication between system components used secure communication mechanisms, while access to dashboards and stored records was controlled through authentication and role-based permissions. Data collection was limited to the purposes stated to participants.

The system also supported configurable **no-capture zones**. This was particularly important for Sri Lankan heritage locations that may contain active religious or culturally sensitive areas. The application could suspend camera-based processing within such zones.

### B.4 Voluntary Participation and Withdrawal

Participation in the pilot was voluntary. Participants were informed that they could withdraw their consent at any time without consequence. Withdrawal could apply to one service or to all services. Once consent for a service was withdrawn, collection for that service was stopped and deletion was initiated according to the project's data-handling policy.

Participants were also informed about the purpose of each monitored service before selecting whether to participate in it. This supported the framework's principle of granular, service-specific consent.

### B.5 Participant Responsibilities

Participants were expected to use the application and relevant services according to the instructions provided by the research team. Participants using driver mode were required to use the vehicle monitoring equipment and follow the defined route while ensuring that the in-vehicle unit and the smartphone were mounted appropriately and that interaction with the device did not require unsafe manual operation while driving.

Participants were also asked to complete the relevant post-visit questionnaire. The questionnaire included ratings of visit experience, perceived accuracy of emotion estimates, usefulness of recommendations, planning quality, trust, privacy, and willingness to use the system again.

### B.6 Ethical Safeguards

The pilot followed the ethical principles specified in the research methodology:

- participation was restricted to adults aged 18 years or above;
- written informed consent was obtained before data collection;
- consent was handled on a per-service basis;
- participation could be withdrawn without consequence;
- raw facial images were not stored or transmitted;
- records were pseudonymized;
- data were collected only for stated research purposes;
- no-capture zones could be configured for culturally or religiously sensitive locations;
- emotion and driver-state outputs were treated as advisory rather than definitive judgments;
- manager-facing information was aggregated to reduce unnecessary exposure of individual data.

These measures were designed to support the requirements of the Sri Lankan Personal Data Protection Act, No. 9 of 2022, together with the project's privacy-by-design principles.

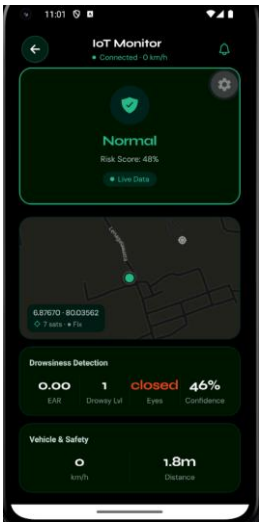
The complete participant information sheet and consent statement are provided in **Appendix L**.

## Appendix C: User interface screens



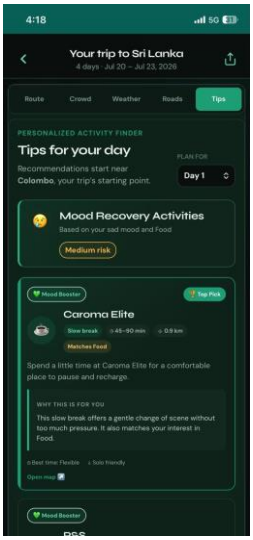






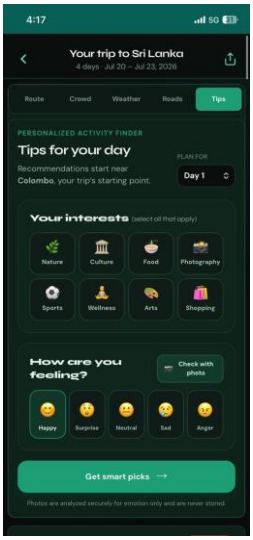


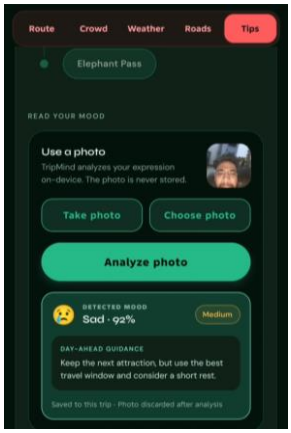


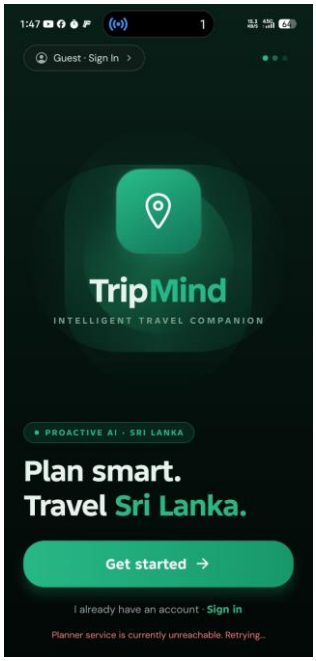


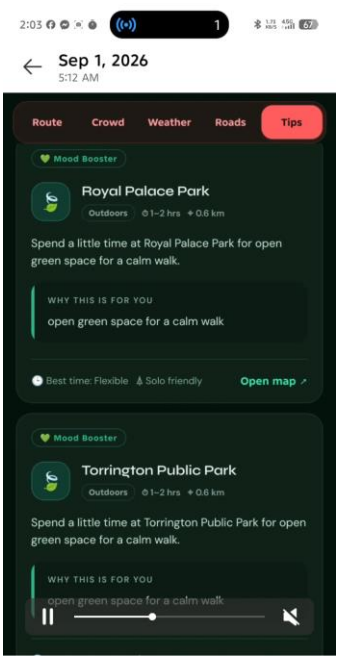


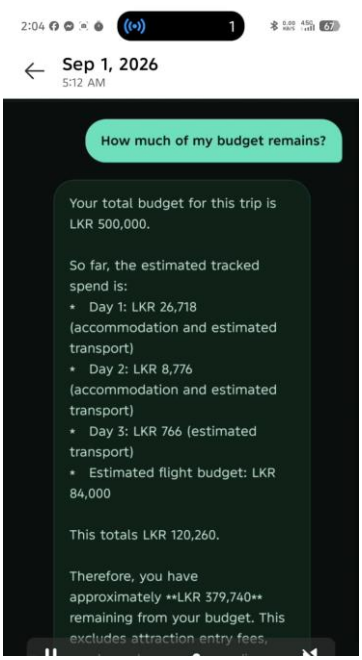


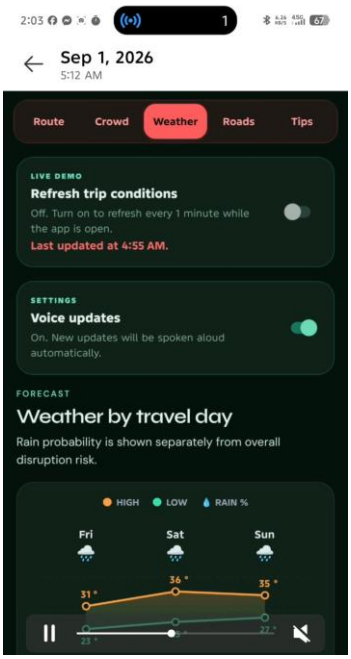


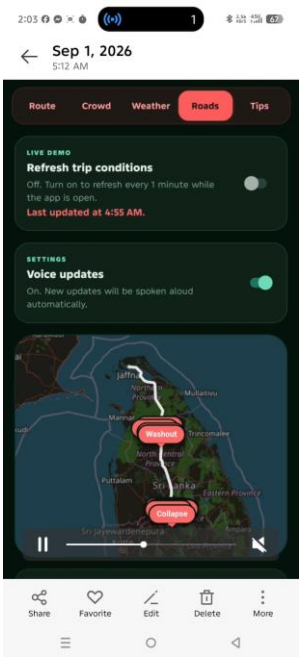



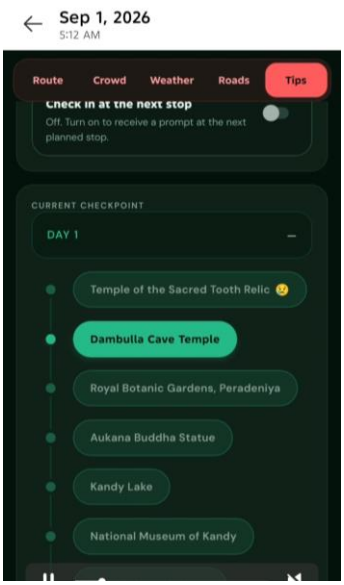

Guest · Sign In
TripMind
INTELLIGENT TRAVEL COMPANION
PROACTIVE AI · SRI LANKA
Plan smart.
Travel Sri Lanka.
Get started
I already have an account · Sign in
Planner service is currently unreachable. Retrying...


Sep 1, 2026
5:12 AM
Route
Crowd
Weather
Roads
Tips
Mood Booster
Royal Palace Park
Outdoors
1–2 hrs
0.6 km
Spend a little time at Royal Palace Park for open green space for a calm walk.
WHY THIS IS FOR YOU
open green space for a calm walk
Best time: Flexible
Solo friendly
Open map
Mood Booster
Torrington Public Park
Outdoors
1–2 hrs
0.6 km
Spend a little time at Torrington Public Park for open green space for a calm walk.
WHY THIS IS FOR YOU
open green space for a calm walk


Sep 1, 2026
5:12 AM
How much of my budget remains?
Your total budget for this trip is LKR 500,000.
So far, the estimated tracked spend is:
Day 1: LKR 26,718 (accommodation and estimated transport)
Day 2: LKR 8,776 (accommodation and estimated transport)
Day 3: LKR 766 (estimated transport)
Estimated flight budget: LKR 84,000
This totals LKR 120,260.
Therefore, you have approximately **LKR 379,740** remaining from your budget. This excludes attraction entry fees.


Sep 1, 2026
5:12 AM
Route
Crowd
Weather
Roads
Tips
LIVE DEMO
Refresh trip conditions
Off. Turn on to refresh every 1 minute while the app is open.
Last updated at 4:55 AM.
SETTINGS
Voice updates
On. New updates will be spoken aloud automatically.
FORECAST
Weather by travel day
Rain probability is shown separately from overall disruption risk.
HIGH
LOW
RAIN %
Fri
Sat
Sun
31 °
36 °
35 °
23 °
27 °


Sep 1, 2026
5:12 AM
Route
Crowd
Weather
Roads
Tips
LIVE DEMO
Refresh trip conditions
Off. Turn on to refresh every 1 minute while the app is open.
Last updated at 4:55 AM.
SETTINGS
Voice updates
On. New updates will be spoken aloud automatically.
Washout
Collapse
Share
Favorite
Edit
Delete
More


Sep 1, 2026
5:12 AM
Route
Crowd
Weather
Roads
Tips
Check in at the next stop
Off. Turn on to receive a prompt at the next planned stop.
CURRENT CHECKPOINT
DAY 1
Temple of the Sacred Tooth Relic
Dambulla Cave Temple
Royal Botanic Gardens, Peradeniya
Aukana Buddha Statue
Kandy Lake
National Museum of Kandy

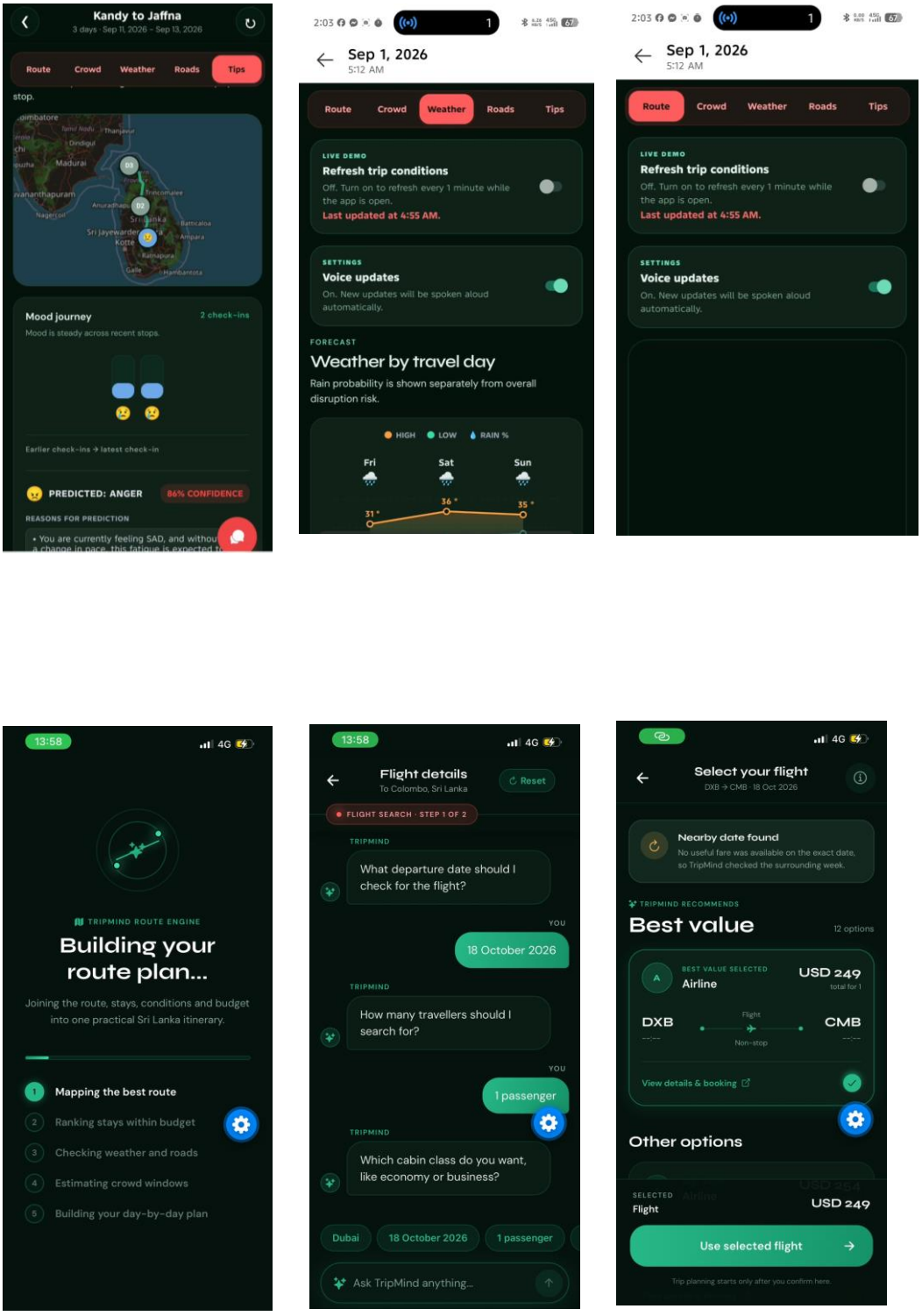



**Figure C.1: User Interface Screens (Original in color)**

## Appendix D: Cross-component event schemas

The following outline gives the common envelope that every message in the framework carries. The complete JSON schemas for each message type are held in the shared contracts repository.

**Table D.1: Common message envelope**

| Field | Type | Description |
|---|---|---|
| schema_version | string | Version of the schema used to validate the message |
| message_type | string | For example ZoneEmotionState or CrowdForecast |
| message_id | UUID | Unique identifier, used for idempotent storage |
| timestamp | ISO 8601 | Time at which the message was produced |
| scope | enum | session or aggregate |
| session_pseudonym | string or null | Random identifier; null for aggregate messages |
| site_id / zone_id / segment_id | string | Place identifiers from the shared spatial model |
| payload | object | Content specific to the message type |
| confidence / sample_count | number | Confidence of the estimate and the number of records behind it |

## Appendix E: Database schema

The principal tables are sessions (pseudonymous identifier, start and end time, consent version and consented services), sites, zones (identifier, type, geometry, amenities, capacity), route segments (identifier, geometry, road class, attributes), emotion records (session, zone, timestamp, fused probabilities, valence, confidence), anomalies (type, severity, evidence, status), driver trip summaries, crowd forecasts, recommendations and their outcomes, and an audit log of cross-component data flows.

## Appendix F: Additional results

This appendix provides supplementary quantitative results from the evaluation of the four framework components, cross-component integration, system performance, and usability. The results presented here are derived from the evaluation procedures described in Chapter 3.

### F.1 Component C1 - Conversational Travel Analysis and Package Optimization

The conversational travel analyzer achieved an **intent classification accuracy of 92%**, exceeding the target of at least 90%. The entity-extraction slot F1 score was reported as **0.88**, exceeding the target of 0.85.

The NSGA-II-based package optimizer produced a hypervolume of **0.82**, compared with **0.67** for the baseline planner. The optimized top-ranked package reduced the mean total cost from approximately **LKR 210,000 to LKR 185,000**, while the mean interest-match score increased from **3.7 to 4.3**.

A total of **87% of requests were resolved without clarification**. The median package-generation time was **6.8 seconds**, while the adaptive engine generated itinerary-repair options in a median of **3.6 seconds** after a triggering event. The user-rated usefulness of generated packages was **4.2/5**, with a standard deviation of **0.6**.

These results indicate that the C1 component achieved its principal planning, optimization, and response-time targets.

### F.2 Component C2 - Adaptive Emotion Recognition

The Custom CNN emotion-recognition model was evaluated on **2,834 RAF-DB test samples** across five emotion classes: anger, happy, neutral, sad, and surprise.

The model correctly classified **2,428 of 2,834 samples**, corresponding to an overall accuracy of **85.67%**. Macro precision was **82.05%**, macro recall was **84.52%**, and macro F1-score was **83.15%**.

| Emotion | Precision | Recall | F1-score |
|---|---|---|---|
| Anger | 72.34% | 83.95% | 77.71% |
| Happy | 95.00% | 89.70% | 92.27% |
| Neutral | 81.59% | 81.47% | 81.53% |
| Sad | 76.10% | 83.26% | 79.52% |
| Surprise | 85.23% | 84.19% | 84.71% |
| **Macro average** | **82.05%** | **84.52%** | **83.15%** |

The highest recall was obtained for the happy class at **89.70%**, while the lowest recall was observed for neutral at **81.47%**. The confusion matrix indicated confusion between visually similar classes. For example, 76 neutral samples were classified as sad, while 42 sad samples were classified as neutral.

The custom CNN achieved 85.67% test accuracy, exceeding the target of at least 80%. The thesis also identifies that the verified INT8 quantization loss and final fusion-ablation results were not available in the current common report. Therefore, these values should not be represented as completed results unless they are confirmed from the final C2 experiment records.

### F.3 Component C3 - Driver and Vehicle Safety

The driver and vehicle safety component achieved high benchmark performance for its lightweight models.

| Evaluation measure | Result | Target |
|---|---|---|
| Eye-state model accuracy – MRL Eye | 99.21% | >90% |
| Yawn model accuracy – YawDD | 99.39% | >90% |
| EAR fatigue trigger logic | Pass | Zero false triggers on blinks |
| Forward-collision TTC tests | Pass with limits | Accurate closing-gap detection |
| Tiered alerts and vibration isolation | Pass | Vibration only at highest risk |
| Privacy verification | Pass | No images off-device; no GPS in FCM alert |

The results demonstrate that the eye-state and yawn models exceeded their benchmark targets. The fatigue-trigger logic and forward-collision testing also passed the defined test procedures, subject to the limitations documented in the main evaluation.

The privacy tests confirmed that raw images remained on the vehicle/device and that GPS information was not included in the Firebase Cloud Messaging alert.

### F.4 Component C4 - Tourism Flow Optimization

The selected C4 forecasting model achieved a **24-hour forecast MAE of 8.4** and **MAPE of 11.8%**, satisfying the target of no more than 15% MAPE.

The seasonal baseline produced an MAE of **12.7** and MAPE of **18.6%**, indicating improved forecasting performance for the selected model.

Prediction interval coverage was **91%**, compared with a nominal target of 87%. Simulation indicated a **14% reduction in peak occupancy** when 20% of visitors accepted redistribution recommendations.

During the pilot, the observed advice acceptance rate was **68%**, with an average additional travel time of **7.5 minutes** for accepted recommendations. The optimizer required **2.8 seconds** per recommendation round.

### F.5 Cross-Component Integration

Three cross-component scenarios were evaluated to determine whether information produced by one component could successfully trigger or influence another component.

| Scenario | Components | Runs | End-to-end completion | Median latency |
|---|---|---|---|---|
| A: Crowd-aware itinerary adaptation | C4 + C1 | 20 | 95% | 3.2 s |
| B: Emotion-triggered plan change | C2 + C1 | 20 | 90% | 3.8 s |
| C: Fatigue-aware transport and visit plan | C3 + C1 + C4 | 20 | 90% | 4.1 s |

Scenario A demonstrated that crowd predictions could trigger itinerary adjustment. Scenario B demonstrated that emotion-related input could result in modification of the

travel plan. Scenario C demonstrated that driver-safety information could influence transport and visit planning.

### F.6 Resilience Testing

The framework was also evaluated when individual components were deliberately made unavailable. C2, C3, and C4 continued to operate when C1 was unavailable. C1 continued to provide planning without emotion, fatigue, or crowd feedback when the corresponding components were unavailable.

The event-bus architecture also provided fallback communication through direct API calls or local queues when required. These tests support the architectural design principle that the four components should remain independently deployable and fail gracefully.

### F.7 Device Performance

Testing on two Android mid-range devices and one iOS device produced the following results:

| Device | Median latency | 95th percentile | Battery/hour |
|---|---|---|---|
| Android mid-range device 1 | 95 ms | 140 ms | 11% |
| Android mid-range device 2 | 102 ms | 148 ms | 12% |
| iOS device | 88 ms | 132 ms | 9% |

All three devices remained below the target of 150 ms latency and 15% battery consumption per hour.

### F.8 Backend Load Testing

| Concurrent sessions | Median response | 95th percentile | Error rate |
|---|---|---|---|
| 100 | 180 ms | 320 ms | 0.5% |
| 500 | 260 ms | 480 ms | 1.2% |
| 1,000 | 390 ms | 720 ms | 2.5% |

The results show that the system supported the design target of at least 500 concurrent sessions, although latency and error rate increased as concurrency increased.

### F.9 Usability Results

| Measure | Participants | Result | Target |
|---|---|---|---|
| Tourist application SUS | 30 | 78 | ≥70 |
| Driver mode SUS | 15 | 75 | ≥70 |
| Manager/operator dashboard SUS | 10 | 81 | ≥70 |
| Usefulness of suggestions | 30 | 4.2 ± 0.6 | - |
| Trust in system | 30 | 4.0 ± 0.7 | - |
| Comfort with data handling | 30 | 4.1 ± 0.6 | - |

All three System Usability Scale results exceeded the target score of 70.

### F.10 Interpretation of Additional Results

Overall, the supplementary results demonstrate that C1, C3, C4, system performance, usability, and cross-component integration achieved their principal reported targets. C2 demonstrated useful five-class emotion-recognition performance The custom CNN achieved 85.67% test accuracy, exceeding the target of at least 80%. The thesis identifies the missing verified results for some C2 trajectory, and anomaly experiments; therefore, those values should be added only after confirmation from the relevant experiment logs.

## Appendix G: Individual Component Summaries

The integrated framework was developed as four complementary components. Each component was assigned to an individual student who was primarily responsible for its design, implementation, evaluation, and documentation. The four components were connected through a shared integration layer consisting of common API contracts, event interfaces, consent and session services, and a shared spatial model.

### G.1 Component C1 - AI Conversational Travel Analyzer, Package Generation and Adaptive Engine

**Responsible student:** Senevirathna T.M.N.D.L (IT22082442)

Component C1 addresses the planning and trip-adaptation stage of the tourist journey. Its primary objective is to convert natural-language travel requests into personalized travel packages and subsequently adapt the accepted itinerary when conditions change.

The component accepts text and voice-based travel requests and extracts important travel constraints such as destinations, dates, budget, group composition, and interests. A conversational approach is used to reduce the need for structured form filling and to support requests containing multiple intents.

The package-generation module uses a multi-objective **NSGA-II optimizer**. The optimization considers competing objectives such as cost, travel time, interest match, comfort, and diversity. Rather than producing only one solution, the optimizer generates non-dominated alternatives from which a suitable package can be selected.

C1 also contains a real-time adaptive engine. When an event such as crowding, a closure, weather-related change, or another relevant condition invalidates part of an accepted itinerary, the adaptive engine generates repair options. A proactive voice agent can communicate the proposed change to the traveler.

The evaluation reported an intent-classification accuracy of **92%**, an NSGA-II hypervolume of **0.82 compared with 0.67 for the baseline**, a reduction in mean package cost from **LKR 210,000 to LKR 185,000**, and an increase in mean interest-match score from **3.7 to 4.3**. The median package-generation time was **6.8 seconds**, while adaptive repair options required **3.6 seconds**.

Within the integrated framework, C1 consumes relevant information from C2, C3, and C4 and publishes itinerary-change events. The component therefore acts as an important decision layer through which experience, safety, and crowd information can influence the tourist's plan.

### G.2 Component C2 - Adaptive Emotion Recognition for Tourist Experience Management

**Responsible student:** Kaweeshwara M.D.O (IT22216014)

Component C2 uses a custom CNN trained on a five-class subset of RAF-DB for facial emotion recognition. OpenCV Haar Cascade is used for face detection, followed by grayscale preprocessing, normalization, and crop-ensemble inference. The component provides zone-wise satisfaction analysis, displays emotion changes over time, and recommends places based on users' selected interests.

The component uses an on-device Custom CNN model to classify five facial-expression categories:

- anger;
- happy;
- neutral;
- sad; and
- surprise.

The model operates on the user's device so that raw facial images do not need to be transmitted to the server.

C2 also models the visitor's emotional state over time rather than treating each classification as an isolated observation. An emotional trajectory is associated with the zones visited by the participant. The system is designed to identify patterns such as sustained negative emotion, abrupt changes, and zone-related dissatisfaction. These signals can be converted into visitor suggestions and manager-facing recommendations.

The benchmark evaluation used **2,834 RAF-DB test samples**. The model achieved **85.67% accuracy**, **82.05% macro precision**, **84.52% macro recall**, and **83.15% macro F1-score**. The highest recall was obtained for the happy class at 89.70%.

The custom CNN achieved 85.67% test accuracy, exceeding the target of at least 80%.

C2 publishes events such as **ZoneEmotionState** and **EmotionAnomaly**, allowing the planning component and other services to use relevant aggregated experience information.

### G.3 Component C3 - Smart Driver and Vehicle Proactive Safety System

**Responsible student:** Rupasinghe M.L.K.S. (IT22176974)

Component C3 addresses the transport stage of the tourist journey. It was designed to provide proactive driver and vehicle safety monitoring without requiring raw visual information to leave the vehicle.

The component uses an edge-based architecture consisting of an **ESP32-CAM**, **HC-SR04 ultrasonic sensor**, ESP32 main hub, and cellular communication module. Lightweight MobileNetV2-based models are used for eye-state and yawn detection. Driver fatigue information is derived using **Eye Aspect Ratio (EAR)**, while forward collision risk is estimated using **Time-to-Collision (TTC)**.

These signals are combined into a **Composite Safety Index (CSI)**. The resulting risk level is used to generate tiered alerts. The alert design includes cabin-level alerts, driver application notifications, and higher-level fleet notifications. Steering-wheel vibration is reserved for the highest risk level.

The component achieved **99.21% eye-state classification accuracy on the MRL Eye dataset** and **99.39% yawn-detection accuracy on the YawDD dataset**. The fatigue-trigger tests and tiered alert tests passed their defined acceptance criteria. Privacy verification also confirmed that raw images remained off-device and that GPS data were not included in the FCM alert.

C3 publishes driver-risk information to the integration layer so that safety information can be considered by other parts of the tourism framework. For example, fatigue-related information can contribute to decisions concerning rest stops or the timing and structure of the wider itinerary.

### G.4 Component C4 - Intelligent Tourism Flow Optimization System

**Responsible student:** Manthrirathne M.A.A.C (IT22607614)

Component C4 addresses visitor-flow management across the destination network. Its objective is to forecast crowding and provide recommendations that redistribute visitors across alternative attractions, times, and routes.

The component uses historical visitor counts together with calendar, weather, and available live signals to forecast future attraction occupancy. Forecasting is followed by an optimization and recommendation process so that the system does not stop at predicting crowding but produces actionable redistribution advice.

The operator dashboard provides information about current and predicted occupancy and supports monitoring and scenario analysis. Recommendations can be reviewed by operators and their projected effects can be assessed.

The selected forecasting model achieved a **24-hour MAE of 8.4 and MAPE of 11.8%**, compared with **12.7 MAE and 18.6% MAPE** for the seasonal baseline. Prediction interval coverage was 91%. Simulation indicated a 14% reduction in peak occupancy when 20% of visitors accepted the recommendations. The observed pilot recommendation acceptance rate was 68%, while the mean additional travel time for accepted recommendations was 7.5 minutes.

C4 publishes **CrowdForecast** and **RedistributionAdvice** events. These outputs can be consumed by C1 so that crowd predictions influence itinerary planning and adaptation.

### G.5 Shared Integration Contribution

Although each student owned an individual component, the overall framework was developed collaboratively. The shared work included the common API gateway, versioned data contracts, event-bus architecture, consent and session services, shared database schema, integration testing, resilience testing, pilot coordination, and preparation of the common report.

The integration layer allows the four components to exchange information while remaining independently deployable. C1 can consume crowd, emotion, and safety information; C2 can provide zone-level experience information; C3 provides driver-risk information; and C4 provides crowd forecasts and redistribution recommendations.

The resulting architecture therefore follows a closed-loop model in which information generated at one stage of the tourist journey can influence decisions at another stage. This is the principal distinction between the integrated framework and isolated single-purpose tourism applications.

## Appendix H: Test case catalogue

The following test cases were executed as part of the layered testing strategy described in Section 2.8.3. The catalogue includes component-level, privacy, performance, integration, and resilience tests. A test is marked **Pass** when the observed result satisfies the defined acceptance criterion and **Fail** when the measured result does not satisfy the target.

**Table H.1: Executed Test Cases**

| ID | Area | Test description | Expected result | Result |
|---|---|---|---|---|
| TC01 | Privacy | Capture network traffic during system operation | No raw facial image data is transmitted | **Pass** |
| TC02 | Privacy | Inspect device storage after an emotion-recognition session | No raw facial images or crops are stored | **Pass** |
| TC03 | C2 Model | Evaluate the custom CNN on the five-class RAF-DB test subset | Accuracy of at least 80% | **Pass - 85.67% accuracy** |
| TC04 | C2 Model | Evaluate emotion classification across the five target classes | Per-class precision and recall are reported for anger, happy, neutral, sad and surprise | **Pass** |
| TC05 | C3 Eye State | Evaluate eye-state classification on the MRL Eye dataset | High classification accuracy is achieved | **Pass – 99.21% accuracy** |
| TC06 | C3 Yawning | Evaluate yawn detection on the YawDD dataset | High classification accuracy is achieved | **Pass – 99.39% accuracy** |
| TC07 | C3 Fatigue | Evaluate EAR-based fatigue triggering using blink sequences | Fatigue alert is triggered correctly without false triggers during normal blinks | **Pass** |
| TC08 | C3 Safety | Evaluate time-to-collision detection against defined safety limits | TTC warnings are generated within the defined limits | **Pass** |
| TC09 | C3 Alerts | Evaluate tiered driver alerts and vibration feedback | Correct alert level and vibration response are produced | **Pass** |

| | | | | |
|---|---|---|---|---|
| TC10 | C3 Privacy | Verify data generated by driver monitoring | No facial images are transmitted off-device and GPS is not included in FCM alerts | **Pass** |
| TC11 | C4 Forecast | Generate tourism-demand forecasts using historical data | Forecast error remains within the defined evaluation target | **Pass – MAE 8.4; MAPE 11.8%** |
| TC12 | C4 Baseline | Compare the proposed forecast model with the seasonal baseline | Proposed model performs better than the baseline | **Pass – baseline MAE 12.7; MAPE 18.6%** |
| TC13 | C4 Prediction Interval | Evaluate prediction interval coverage | Observed coverage is close to or above the required nominal coverage | **Pass – 91% coverage** |
| TC14 | C4 Advice | Simulate peak tourism occupancy and generate alternative advice | System produces advice that can reduce peak occupancy | **Pass – simulated reduction 14%** |
| TC15 | C4 Advice | Evaluate visitor acceptance of generated advice | A majority of visitors accept the suggested alternatives | **Pass – 68% acceptance** |
| TC16 | C4 Optimization | Measure optimization execution time | Optimization completes within practical response time | **Pass – 2.8 s** |
| TC17 | System Performance | Measure mobile inference/response latency on Android device 1 | Response remains within the defined latency target | **Pass – median 95 ms; p95 140 ms** |
| TC18 | System Performance | Measure mobile inference/response latency on Android device 2 | Response remains within the defined latency target | **Pass – median 102 ms; p95 148 ms** |
| TC19 | System Performance | Measure mobile inference/response latency on iOS | Response remains within the defined latency target | **Pass – median 88 ms; p95 132 ms** |
| TC20 | Load | Execute 100 concurrent sessions | System remains operational with | **Pass – 180 ms average; 320 ms p95; 0.5% error** |

| | | | | |
|---|---|---|---|---|
| | | | acceptable latency and error rate | |
| TC21 | Load | Execute 500 concurrent sessions | System remains operational with acceptable latency and error rate | **Pass – 260 ms average; 480 ms p95; 1.2% error** |
| TC22 | Load | Execute 1,000 concurrent sessions | System remains operational within the defined load limits | **Pass – 390 ms average; 720 ms p95; 2.5% error** |
| TC23 | Battery | Measure one-hour active operation on Android device 1 | Battery consumption remains at or below 15% per hour | **Pass – 11%/h** |
| TC24 | Battery | Measure one-hour active operation on Android device 2 | Battery consumption remains at or below 15% per hour | **Pass – 12%/h** |
| TC25 | Battery | Measure one-hour active operation on iOS | Battery consumption remains at or below 15% per hour | **Pass – 9%/h** |
| TC26 | INT-A | Trigger a crowd forecast above threshold with an affected itinerary | C4 provides information that allows C1 to generate an itinerary adaptation | **Pass – 95% end-to-end completion; 3.2 s median latency** |
| TC27 | INT-B | Trigger emotion-related dissatisfaction affecting itinerary planning | C2 information is successfully passed to C1 for adaptive itinerary processing | **Pass – 90% end-to-end completion; 3.8 s median latency** |
| TC28 | INT-C | Trigger driver fatigue before a tourism-flow risk condition | C3, C1 and C4 produce a coordinated response | **Pass – 90% end-to-end completion; 4.1 s median latency** |
| TC29 | INT Resilience | Make one component unavailable during an integrated session | Remaining components continue operating with reduced functionality | **Pass** |
| TC30 | Usability | Evaluate the tourist application using the System Usability Scale | System achieves acceptable usability | **Pass – SUS 78** |

| | | | | |
|---|---|---|---|---|
| TC31 | Usability | Evaluate driver mode using the System Usability Scale | System achieves acceptable usability | **Pass – SUS 75** |
| TC32 | Usability | Evaluate manager/operator dashboards using the System Usability Scale | System achieves acceptable usability | **Pass – SUS 81** |
| TC33 | User Evaluation | Evaluate perceived usefulness of the system | Users report positive perceived usefulness | **Pass – 4.2/5** |
| TC34 | User Evaluation | Evaluate user trust in the system | Users report positive trust levels | **Pass – 4.0/5** |
| TC35 | User Evaluation | Evaluate comfort with data handling | Users report acceptable comfort with data handling | **Pass – 4.1/5** |

## Appendix I: System Usability Scale questionnaire

Participants rate each statement from 1 (strongly disagree) to 5 (strongly agree) [25]. Odd-numbered items are positively worded and even-numbered items negatively worded.

**Table I.1: System Usability Scale items**

| No. | Statement |
|---|---|
| 1 | I think that I would like to use this system frequently. |
| 2 | I found the system unnecessarily complex. |
| 3 | I thought the system was easy to use. |
| 4 | I think that I would need the support of a technical person to be able to use this system. |
| 5 | I found the various functions in this system were well integrated. |
| 6 | I thought there was too much inconsistency in this system. |
| 7 | I would imagine that most people would learn to use this system very quickly. |
| 8 | I found the system very cumbersome to use. |
| 9 | I felt very confident using the system. |
| 10 | I needed to learn a lot of things before I could get going with this system. |

**Scoring.** For odd items the contribution is the response minus 1. For even items the contribution is 5 minus the response. The sum of contributions is multiplied by 2.5 to give a score from 0 to 100. A score of 68 is commonly taken as average, and the project target is at least 70. The score is reported separately for the tourist application, driver mode and each dashboard, with the number of participants and the standard deviation.

## Appendix J: Post-visit questionnaire

The questionnaire collects the self-reported reference for emotion estimates and the perceived value of the system. Items use a five-point scale unless stated.

**Table J.1: Post-visit questionnaire items**

| Section | Item | Response type |
|---|---|---|
| Visit experience | For each zone visited, how did you feel overall? (very negative to very positive) | Five-point scale per zone |
| Visit experience | For each zone, which of these best describes how you felt? (happy, neutral, surprised, sad, angry, disgusted, fearful, bored) | Single choice per zone |
| Visit experience | Was there a moment when you felt uncomfortable, frustrated or unsafe? Where and why? | Free text |
| Accuracy | The emotion shown by the app matched how I felt. | Five-point agreement |
| Usefulness | The suggestions given by the app were useful. | Five-point agreement |
| Usefulness | I followed at least one suggestion. If so, what happened? | Yes or no; free text |
| Planning | The planner understood what I asked for. | Five-point agreement |
| Planning | The changes proposed to my itinerary were sensible. | Five-point agreement |
| Trust and privacy | I understood what data the app collected. | Five-point agreement |
| Trust and privacy | I felt comfortable with the camera being used in the way described. | Five-point agreement |
| Trust and privacy | I would use this app again on a future visit. | Five-point agreement |
| Background | Age group, country, first visit to the site (yes or no), phone model | Selection |

| Section | Item | Response type |
|---|---|---|
| Open feedback | What would you change or add? | Free text |

**Appendix K: Interview guide for managers and operators**

Semi-structured interviews of about 30 minutes are held with managers, operators and, where available, site staff and drivers. The following questions guide the interview and are adapted to the role.

1. Please describe how you currently learn about visitor satisfaction and crowding at your site or in your fleet.
2. How quickly do you currently learn about a problem, and what do you do about it?
3. After using the dashboard, which information did you find most useful, and what was missing?
4. Were the alerts and recommendations clear? Was the evidence sufficient for you to act?
5. How often did you receive alerts, and was this too many, too few or about right?
6. What concerns would your staff, drivers or visitors have about this kind of system?
7. What thresholds or settings would you change for your site, and why?
8. What would be needed for your organization to adopt a system like this?
9. Is there anything about the data handling that would need to change for you to be comfortable using it?

## Appendix L: Participant information and consent text

### Participant information

You are invited to take part in a research study carried out by students of the Sri Lanka Institute of Information Technology (SLIIT). The study evaluates a smartphone application that helps to plan trips, estimates visitor emotions at heritage sites, monitors driving safety, and forecasts tourism crowding.

Participation in this study is voluntary. You may choose not to participate or withdraw from the study at any time without providing a reason and without any negative consequences.

If you agree to participate, you will install and use the application during your visit. Depending on the services you choose, the application may use the front camera to estimate facial expressions, location information to determine the general area of the site you are visiting, vehicle sensors for driver-safety monitoring, and spoken or written travel requests for trip planning.

**Photographs and video are not stored and are not sent from your phone.** The phone processes the camera input to generate numerical emotion-related features. Only the required numerical features, the general area of the site, the time, and a randomly generated code that is not linked to your name may be transmitted. Your exact location is not transmitted as part of the emotion-recognition data.

The data collected during the study are used for research purposes. Appropriate security and encryption measures are applied to protect the data, and access is limited to authorized members of the research team. Data may be retained for a maximum period of 12 months. Research results will be reported in aggregate form, and individual participants will not be identified.

You may request deletion of your research data during the data-retention period, subject to applicable research requirements. Emotion estimates produced by the application are approximate and are **not a medical, psychological, or clinical assessment**.

**Consent Statement**

Please tick the services you agree to participate in using the consent selections in **Table L.1** and sign below.

I confirm that I am **18 years of age or older**, that I have read and understood the information provided above, that I have had the opportunity to ask questions, and that I understand that I can withdraw from the study at any time without providing a reason.

**Name:** _____

**Signature:** _____

**Date:** _____

**Appendix N: Example messages**

The following examples illustrate the common envelope described in Appendix D. Values are illustrative.

***Emotional state record uploaded by C2***

{ "schema_version": "1.0", "message_type": "EmotionRecord", "message_id": "9f2c…", "timestamp": "2026-09-14T09:42:10+05:30", "scope": "session", "session_pseudonym": "a41e…", "zone_id": "SIG-CLIMB-02", "payload": { "probabilities": [0.05, 0.02, 0.23, 0.10, 0.60], "valence": -1, "low_confidence": false }, "confidence": 0.60 }

***Zone state event published by C2***

{ "schema_version": "1.0", "message_type": "ZoneEmotionState", "timestamp": "2026-09-14T09:45:00+05:30", "scope": "aggregate", "site_id": "SIG", "zone_id": "SIG-CLIMB-02", "payload": { "satisfaction": 0.38, "valence_trend": -0.4, "window_minutes": 10 }, "sample_count": 27 }

***Redistribution advice published by C4***

{ "schema_version": "1.0", "message_type": "RedistributionAdvice", "timestamp": "2026-09-14T06:00:00+05:30", "scope": "aggregate", "site_id": "SIG", "payload": { "action": "shift_earlier", "slot": "10:00-11:00", "alternative": "07:00-08:00", "reason": "forecast occupancy above comfort threshold", "projected_peak_reduction": 0.12 }, "confidence": 0.7 }

***Driver risk alert published by C3***

{ "schema_version": "1.0", "message_type": "DriverRiskAlert", "timestamp": "2026-09-14T16:20:00+05:30", "scope": "session", "session_pseudonym": "c7b0…", "segment_id": "SEG-KANDY-NE-05", "payload": { "level": "caution", "cause": "rising_fatigue_before_hill_segment", "recommendation": "rest_stop_within_10_minutes" } }

**Appendix O: Configuration and deployment notes**

The framework is packaged as containers. Each component has its own image and configuration file, and an orchestration file starts the shared services (database, message broker, API gateway) and all components for development and for the pilot. The following steps summaries a deployment.

1. Provision a managed PostgreSQL instance with the PostGIS extension and apply the migration scripts from the contracts repository.
2. Provision a message broker and create the topics for the events listed in Table 2.2.
3. Store secrets (database credentials, signing keys, third-party API keys) in the secret manager and mount them as environment variables.
4. Deploy the API gateway and the component services, and configure TLS 1.3 certificates and the allowed origins for the dashboards.
5. Load the site configuration: zone polygons, zone types, capacities, thresholds and no-capture zones.
6. Register model versions in the model registry and set the active version for each model.
7. Run the smoke tests and the contract tests against the deployed environment.
8. Build and distribute the mobile application to test devices and verify the consent flow before the pilot.

Monitoring covers service health, request latency, error rates, queue depth, model confidence distributions and the rate of alerts per hour. Dashboards for these metrics are used during the pilot, and alerts to the team are raised if the error rate or the queue depth exceeds a limit.

**Appendix P Component methodology and detailed design**

This section summarizes the method of each component. Component C2 is described in more detail because its design is documented in full in the corresponding dissertation and is used as the worked example of how a component is built to the framework’s conventions. The descriptions of C1, C3 and C4 are written at the level needed to

understand the integration and are marked where the component owner must confirm the final parameters before submission.

### Component C1: AI conversational travel analyzer, package generation and adaptive engine

#### *Multi-intent natural language analyzer*

The analyzer receives a request as text. A transformer-based large language model [6], Google Gemini accessed through the LangChain ChatGoogleGenerativeAI interface, is used without task-specific fine-tuning. System prompts instruct it to ignore conversational chit-chat and to return the extracted fields as a structured JSON object that follows a fixed schema. Intake is split into two phases. In the flight phase the model extracts the departure city, departure date, number of passengers, cabin class and total budget; city names are mapped to airport codes through an alias table and dates are standardized to the ISO format. In the trip phase it extracts the local starting point, the destination and the duration, and the request is marked complete only when all three are present. At every turn the assistant checks the session state and asks a short clarifying question for any missing field, so that no partial request is sent to the flight, routing or planning services. The two-phase design was adopted after a single combined chat repeatedly confused the international departure city with the domestic starting point, and changes to the prompt alone did not remove the error.

#### *Package optimizer*

Package generation is formulated as a multi-objective optimization problem. A candidate package is a sequence of sites, accommodation and transport segments over the requested days. The objectives are to minimize total cost, to minimize travel time, to maximize the match between the package and the stated interests, and to maximize comfort and diversity, subject to constraints such as the budget cap, opening hours and the daily travel limit. NSGA-II [7] is used to find a set of non-dominated packages. Candidates are ranked by weights derived from the priorities the user expressed in the conversation, and the top packages are presented with an explanation of the trade-offs, for example “this option is 12 percent cheaper but adds 90 minutes of driving”.

***Route ranking and rule-based selection***

The same NSGA-II [7] is applied to the route alternatives returned by the Google Routes API between the start and the destination, for example an expressway, an inner-country road and a coastal road. Five objectives are minimized: travel time; budget, which is the fuel cost derived from the distance plus expressway tolls; road risk, derived from active road incidents on the route; weather risk; and crowd exposure. A route dominates another if it is not worse in any of the five objectives and strictly better in at least one. The routes are sorted into Pareto fronts by repeated non-dominated sorting, and the crowding distance, which measures how close a route is to its neighbors along each objective, is used within a front to prefer routes in less crowded regions of the objective space and so keep a diverse set of options. The primary route is taken from the first front. Because the candidates are the small set of alternatives returned by the routing service, this ranking uses the selection mechanics of NSGA-II without a population, generations, crossover or mutation. Attractions and accommodation are then chosen by rules. Attractions must lie within 10 km of the route, weak places such as junctions, clock towers and car parks are discarded, and outdoor places are deprioritized when the forecast rain probability exceeds 65 percent. The lodging budget per night is the total budget minus the flight cost, divided by the number of days; hotels above this budget or more than 5 km from the end point of the day are removed, and the top three of the remainder are recommended by a weighted score of rating, price and amenities.

***Real-time adaptive engine and proactive voice agent***

Once a package is accepted it is monitored by the adaptive engine. The engine receives conditions as events: weather, closures and delays, crowd forecasts and advice from C4, route-risk and driver alerts from C3, and zone emotion states and anomalies from C2. For each event it evaluates whether the current plan is still acceptable using the user's constraints and the objectives of the optimizer, and if not it generates a small number of repair options by re-running a reduced version of the optimizer with the already completed parts of the plan fixed. The proactive voice agent delivers the repair options by speech and text, allows the user to accept or decline by voice, and respects quiet periods and the rate limits in Section 2.3.5.

**Component C2: Adaptive emotion recognition for tourist experience management**

***Architecture***

Component C2 follows four layers, illustrated in Figure P.1. **Layer 1 (mobile capture)** provides the React Native interface for capturing check-in images and obtaining timestamps and location information. **Layer 2 (emotion recognition)** detects faces using OpenCV Haar Cascade and applies grayscale preprocessing, normalization, and custom CNN inference to produce an emotion label and confidence. **Layer 3 (data management and analysis)** stores emotion records with timestamps and zone identifiers, summarizes zone-wise satisfaction, and organizes records to track emotion changes over time. **Layer 4 (output and recommendations)** displays zone satisfaction and emotion trajectories on the manager dashboard and suggests places to tourists based on their selected interests.

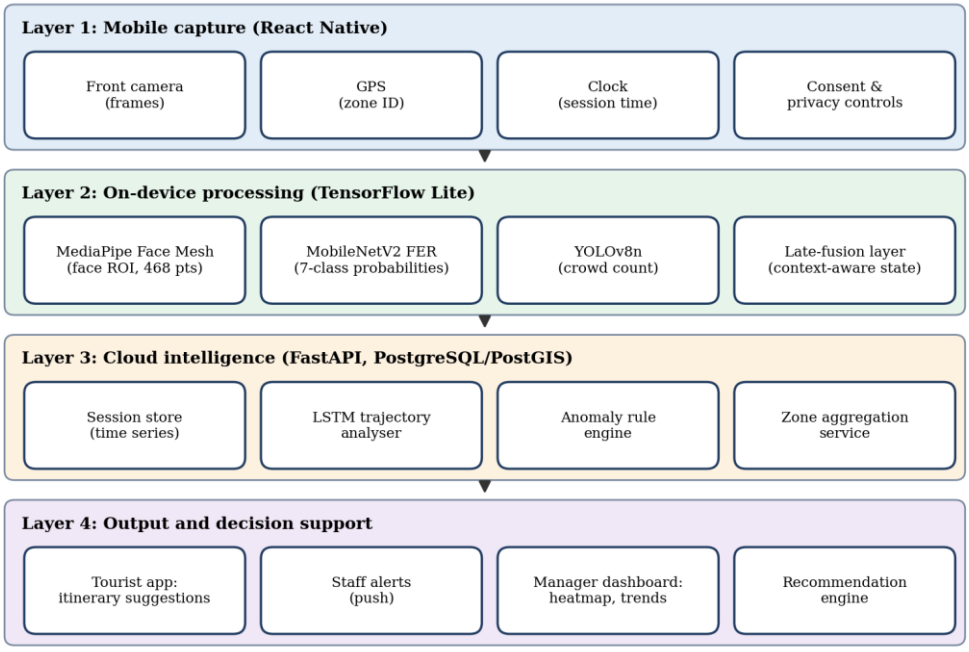



**Figure P.2 : Four-layer architecture of the adaptive emotion recognition component (C2)**

### *Sampling strategy*

Emotion recognition is initiated when the user captures a check-in image. OpenCV Haar Cascade detects the largest face, with −15° and +15° rotation fallback when initial detection fails. The facial image is resized to 100 × 100 pixels, converted to grayscale, and normalized. The custom CNN performs inference using crop scales of 0.95, 1.0, and 1.05, and the resulting probability vectors are averaged to obtain the final emotion label and confidence. Emotion trajectories therefore represent recorded check-ins rather than continuous monitoring.

### *Facial emotion recognition module*

The facial emotion recognition model was developed using a custom Deep Convolutional Neural Network (CNN) trained on the RAF-DB dataset. The architecture consists of multiple Conv2D layers for extracting facial features, MaxPool2D layers for reducing spatial dimensions, dropout layers for regularization, and dense fully connected layers for final classification.

Before CNN inference, facial regions are detected using an OpenCV Haar Cascade classifier. The largest detected face is selected and cropped from the input image. To improve robustness to tilted faces, the detection pipeline also attempts rotations of −15° and +15° when a suitable face is not detected in the original image.

The detected facial crop is resized to 100 × 100 pixels, converted to grayscale, and normalized to the [0, 1] range. These preprocessing steps produce the input representation expected by the custom CNN.

To improve prediction robustness, a crop-ensemble inference strategy is used. Three forward passes are performed using crop scaling factors of 0.95, 1.0, and 1.05. The resulting probability vectors are averaged arithmetically to obtain the final emotion probability distribution and predicted emotion class.

The prepared five-class RAF-DB dataset contained 14,107 facial images. The custom CNN was trained on 10,147 images, with 1,126 images used for validation during training. The remaining 2,834 images were reserved for independent testing. The trained model was subsequently exported to TensorFlow Lite format for inference..

The main training settings were:

| Parameter | Value |
|---|---|
| Model | Custom Deep CNN |
| Dataset | RAF-DB |
| Emotion Classes | 5 |
| Architecture | Conv2D + MaxPool2D + Dropout + Dense |
| Face Detection | OpenCV Haar Cascade |
| Maximum epochs | 50 |
| Model Format | TensorFlow Lite (.tflite) |
| Loss function | Categorical cross-entropy |
| Validation Accuracy | Approximately 82–85% |

The model was trained using the training set and validated after each epoch. Early stopping was used to reduce overfitting. The final model was then evaluated using the independent test set.

The custom CNN achieved 85.67% test accuracy on 2,834 samples from the five-class RAF-DB test subset.

***Anomaly detection and decision support***

The anomaly layer combines transparent rules with the trajectory classifier. Rule A (sustained negative emotion) fires when smoothed valence is at or below -1 for longer than a configurable duration within one zone (default three minutes). Rule B (abrupt valence change) fires when valence drops by two or more levels within 30 seconds, which may indicate a safety event. Rule C (zone dissatisfaction) fires when the aggregate satisfaction score of a zone falls below a baseline. an early-warning level is raised before any rule fires [19]. Alerts are rate-limited and de-duplicated, and each records its evidence.

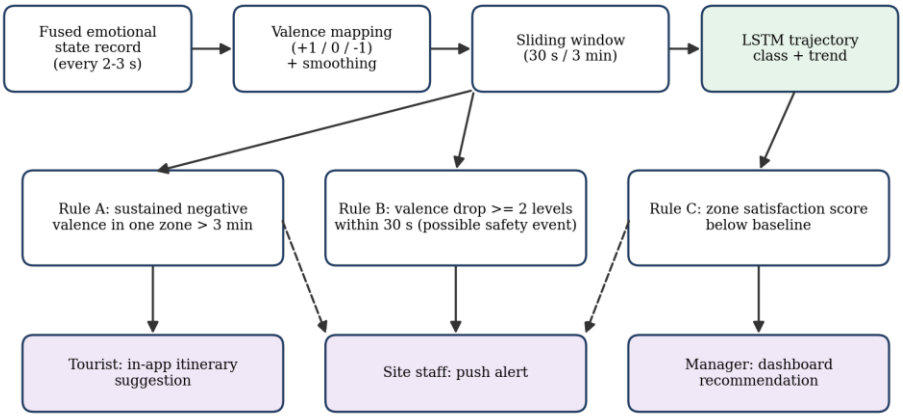



**Figure P.3 : Trajectory analysis and anomaly detection flow in component C2**

The decision support module converts anomalies and aggregated data into two kinds of output. For tourists, it selects suggestions from a zone knowledge base (zone type, amenities, capacity, typical dwell time and neighboring zones), ranked by distance, crowd density, recent satisfaction and fit with the detected issue. For managers, it ranks zones by an impact score that multiplies visitor volume by the proportion of negative or anomalous records and attaches templated recommendations with their evidence. Suggestions are advisory and the visitor remains free to ignore them.

***In-app check-in classifier and mood prediction***

In the mobile application, a check-in selfie is also classified by a convolutional network served by a separate emotion service. The network has several Conv2D and max-pooling layers with dropout and dense layers, is trained on RAF-DB [27] and is exported to TensorFlow Lite; it reaches about 85.67% test accuracy on the five-class RAF-DB test subset. Faces are located with an OpenCV Haar cascade, and if none is found the image is rotated by −15 and +15 degrees and the search is repeated. The largest face is cropped, resized to 100 × 100 pixels, converted to grayscale and scaled to the range 0 to 1. The final prediction is the mean of the probability vectors from three forward passes with crop scales of 0.95, 1.0 and 1.05. The detected emotion is mapped to place profiles, for example parks, gardens and waterfalls for anger, and museums, historic sites and cafes for a neutral mood, and places of those categories within a

default radius of 3 km are retrieved from OpenStreetMap through the Overpass API and filtered by the hobbies of the user.

To avoid travel fatigue and notification overload, the application also runs an on-device first-order Markov predictor of the mood expected at the next attraction. Past check-ins, loaded from the database at the start of a session, are scanned for transitions from the current mood to another mood while visiting the category of the next attraction, and the counts form a transition matrix. Moods are scored excited 5, happy 4, neutral 3, angry 2 and sad 1, and the expected score is the transition-weighted mean of the destination scores; when no matching history exists, the current score decays by 0.5 with a floor of 1. Penalties are then added: −0.3 when the crowd score exceeds 50 and −0.8 when it exceeds 70, and 0-.5 for forecast rain or storm and -0.4 for temperatures above 85 °F. The final score is mapped back to a mood (up to 1.5 angry, up to 2.5 sad, up to 3.5 neutral, up to 4.5 happy, above 4.5 excited). If it is below 2.8, the application warns of fatigue and suggests a rest or a low-intensity indoor alternative.

**Component C3: Smart driver and vehicle proactive safety system**

***IoT Edge Architecture and On-Device Vision***

The system operates as a self-contained IoT unit. An ESP32-CAM equipped with an OV2640 camera captures the driver's face, extracting 68 facial key points to compute the Eye Aspect Ratio (EAR). It runs MobileNetV2 models locally one for eye state (open/closed) and one for yawning (yawn/no yawn). A fatigue state triggers when EAR drops below a calibrated threshold for a specific number of consecutive frames. The camera node outputs only numerical status data; no raw video or images are stored or transmitted.

***Collision Risk and Sensor Fusion***

An HC-SR04 ultrasonic sensor captures the forward distance to the vehicle ahead. This distance, along with the driver status, is transmitted via connectionless ESP-NOW to an ESP32 main hub. The hub computes the relative closing speed and the Time-to-

Collision (TTC). It fuses the visual fatigue metric and the TTC into a single Composite Safety Index (CSI).

***Tiered Alerts and Cloud Connectivity***

The CSI is mapped to escalating risk tiers. The hub, utilizing a SIM7600CE LTE/GNSS module, syncs telemetry to a Firebase Realtime Database for the driver's mobile app (dashboard and live map). Alerts range from an in-cabin buzzer and dashboard warnings to Firebase Cloud Messaging (FCM) push notifications directed to the fleet administrator. FCM notifications carry only the alert type and timestamp to preserve privacy, excluding GPS data. The highest tier of physical intervention steering wheel vibration is strictly reserved for critical threat levels.

**Component C4: Intelligent tourism flow optimization system**

***Crowd forecasting***

The forecasting module predicts occupancy for each attraction and hour for the coming days. Inputs include historical visitor counts, calendar features (weekday, public and school holidays, religious events and festivals), weather forecasts, planned itineraries from C1, and recent live signals such as the number of active sessions in a zone and the density estimates reported by C2. Candidate models include a seasonal baseline, gradient-boosted trees and an LSTM [19]. Models are evaluated by rolling-origin validation with mean absolute error and mean absolute percentage error, and by whether the prediction interval covers the actual value. Where history is short, as it is for a new site, the model uses data from similar sites and widens its uncertainty accordingly. The LSTM model is selected as the primary crowd forecasting model, while a seasonal baseline and gradient-boosted trees are used for comparison. The system forecasts attraction occupancy at an hourly resolution for the next 24 hours.

***Crowd score***

For each attraction and each day of a trip, a crowd score from 0 to 100 is computed as the sum of four signals: a macro seasonality weight for the month, taken from the monthly arrival statistics of the Sri Lanka Tourism Development Authority (SLTDA); a micro interest trend, taken from the monthly Wikipedia pageviews of the article of

each attraction through the Wikimedia Pageviews API and stored in the attractions dataset; a calendar penalty, with 22 points added on public holidays from the Nager. Date service and 10 points on weekends; and a weather risk adjustment from the forecast. When the score exceeds 80, high-crowd places are moved to the earliest morning slot or to the next morning and a warning is added to the daily briefing. The score is also used in the route ranking and in the mood predictor of C2.

***Optimization and recommendation***

When forecast occupancy exceeds a comfort threshold, the optimization module searches for redistribution actions. Candidate actions include shifting a visit to an earlier or later time, recommending a nearby alternative with similar interest tags and lower expected crowding, and suggesting a different order of sites on a route. Each candidate is scored by the reduction in peak occupancy, the inconvenience to the tourist (extra travel time and deviation from stated interests) and the likelihood of acceptance. The module outputs a small ranked set of advice items with reasons, which C1 and the tourist app present to the user. The loop is closed by recording whether advice was accepted and what happened to occupancy afterwards.

***Operator dashboard***

The dashboard shows current and forecast occupancy against capacity for each site, the status of recommendations, scenario analysis (what happens to peak crowding if a given share of visitors accept the advice), and historical trends. Operators can configure capacities and thresholds, which are published as Zone Configuration events.

This section records the design in more detail than the overview in Sections 2.2 to 2.5, so that the framework can be reproduced and so that the examiners and future developers can trace requirements to design. It covers use cases, user stories, interface descriptions, the data model, configuration parameters, key algorithms, scenario sequences and quality assurance practice.

**Use cases**

Table 15 lists the principal use cases of the framework, the actors involved and a summary of the main flow. Use cases UC1 to UC4 belong mainly to C1, UC5 to UC8

to C2, UC9 to UC11 to C3, UC12 to UC14 to C4 and UC15 to UC18 to the shared layer.

**Table P.1: Principal use cases**

| ID | Use case | Actor | Summary of main flow |
|---|---|---|---|
| UC1 | Request a trip by text or voice | Tourist | The tourist enters or speaks a request; the analyzer extracts intents and constraints; the dialogue manager asks clarifying questions for missing items. |
| UC2 | Compare generated packages | Tourist | The system shows ranked packages with cost, time and interest match and explains the trade-offs; the tourist selects one. |
| UC3 | Receive a repaired itinerary | Tourist | An event invalidates part of the plan; the adaptive engine generates options; the voice agent presents them; the tourist accepts or declines. |
| UC4 | Refine preferences in dialogue | Tourist | The tourist changes budget, dates or interests mid-conversation; the system re-optimizes and highlights what changed. |
| UC5 | Give consent and start a session | Tourist | The tourist reads the consent screen, selects services and accepts; a session pseudonym and token are issued. |
| UC6 | Capture emotion during a visit | Tourist | The user captures a check-in image. The custom CNN classifies the detected face, and the resulting emotion record is associated with a timestamp and zone. |
| UC7 | Receive a suggestion on anomaly | Tourist | The anomaly layer fires; the decision module ranks nearby options; a suggestion appears; the tourist may follow or ignore it. |
| UC8 | View journey summary | Tourist | After the visit the app plots the emotional trajectory by zone and lists recommendations. |
| UC9 | Start a monitored trip | Driver | The driver ensures the IoT unit is powered; the ESP32-CAM and ultrasonic sensor begin monitoring fatigue and forward distance. |

| ID | Use case | Actor | Summary of main flow |
|---|---|---|---|
| UC10 | Receive a proactive safety alert | Driver | CSI crosses a risk threshold; a tiered alert (buzzer, dashboard warning, or steering wheel vibration) is triggered; FCM push notifications reach the fleet administrator. |
| UC11 | Review trip safety summary | Driver, Operator | After the trip the score, events and recommendations are shown to the driver and, in summarized form, to the operator. |
| UC12 | Monitor occupancy and forecasts | Operator, Manager | The dashboard shows current and forecast occupancy against capacity for each site and hour. |
| UC13 | Review redistribution advice | Operator | The operator inspects advice items, their reasons and projected effects, and approves or rejects them. |
| UC14 | Run a what-if scenario | Operator | The operator specifies an acceptance rate or capacity change; the system simulates the effect on peak occupancy. |
| UC15 | Configure thresholds and no-capture zones | Manager | The manager edits anomaly thresholds, capacities and no-capture polygons; the changes are published and applied. |
| UC16 | Respond to a safety alert | Site staff | Staff receive a push alert with evidence and zone; they act and mark the alert as handled. |
| UC17 | Withdraw consent | Tourist, Driver | The user withdraws consent for one or all services; collection stops and deletion is triggered according to policy. |
| UC18 | Export anonymized statistics | Authority, Analyst | An authorized analyst exports aggregated statistics with the export recorded in the audit log. |

**User stories and acceptance criteria**

User stories were used to priorities development in each sprint. Table 16 lists representative stories with their acceptance criteria. Priorities follow the MoSCoW scheme (Must, Should, Could).

**Table P.2:Representative user stories**

| ID | User story | Acceptance criterion | Priority |
|---|---|---|---|
| US1 | As a tourist, I want to describe my trip in my own words so that I do not fill in long forms. | At least 90 percent of required slots are filled from one free-text message in the test set, or a clarifying question is asked. | Must |
| US2 | As a tourist, I want to see why a package is recommended so that I can trust it. | Each package shows its cost, time and interest-match score and a one-sentence trade-off explanation. | Must |
| US3 | As a tourist, I want my plan to change when something goes wrong so that I do not lose my day. | After a simulated closure the engine offers at least one valid alternative within the target time. | Must |
| US4 | As a tourist, I want to choose which sensors are used so that I keep control of my data. | The consent screen lets the user select each service independently; unselected services never run. | Must |
| US5 | As a tourist, I want a suggestion when a part of the visit is going badly so that I can feel better. | A suggestion appears within one minute of an anomaly and names a nearby alternative. | Should |
| US6 | As a tourist, I want a summary of my visit so that I can remember it. | The summary shows an emotion trajectory per zone for the completed session. | Could |
| US7 | As a driver, I want warnings before I become dangerously tired so that I can rest. | An alert is issued when the drowsiness score crosses the threshold, with at most the target false alarms per hour. | Must |
| US8 | As a driver, I want clear, non-distracting physical alerts so I can keep my eyes on the road. | The system uses an in-cabin buzzer for early warnings and reserves steering wheel vibration strictly for the highest critical risk. | Must |

| ID | User story | Acceptance criterion | Priority |
|---|---|---|---|
| US9 | As an operator, I want to know about urgent risks immediately without invading driver privacy. | Fleet administrators receive real-time FCM push notifications containing only the alert type and timestamp (no video or GPS). | Should |
| US10 | As a manager, I want to see which zones are unhappy so that I can fix them. | The heatmap shows zone satisfaction with the number of records behind each value. | Must |
| US11 | As a manager, I want recommendations with evidence so that I can justify my decisions. | Each recommendation lists the time window, the share of negative records and the crowd density. | Should |
| US12 | As an operator, I want a forecast for tomorrow so that I can plan staffing. | A forecast by site and hour is available for the next 24 hours with an uncertainty range. | Must |
| US13 | As an operator, I want to simulate the effect of advice so that I can decide whether to apply it. | The scenario tool returns the expected change in peak occupancy for a given acceptance rate. | Could |
| US14 | As site staff, I want only relevant alerts so that I am not overwhelmed. | Alerts are de-duplicated and rate-limited; each shows its evidence. | Must |
| US15 | As a researcher, I want anonymized data so that I can analyses patterns without identifying people. | Exports contain no pseudonyms that can be linked to a person and include sample counts. | Should |

**User interface design**

The interface design follows three rules. Information that the user needs at a glance is shown first. Anything that could distract a driver is delivered by voice or, for C3, by a short in-cabin buzzer tone. Every automated output explains itself in one sentence. The principal screens are described below.

- **Home and consent screen.** Shows the site map, the current zone, the services in use and a prominent control to pause or stop sensing. The first launch shows the consent flow with one toggle per service, a short plain-language explanation of each and a link to the full text.

- **Planner screen (C1).** A conversational view with a text box and a microphone button. Below the conversation, packages are shown as cards with cost, duration and interest-match bars, and an expandable explanation of the trade-offs.
- **Itinerary screen (C1).** A timeline of the day with each stop, travel time and a badge when the engine has changed an item. A banner presents repair options with Accept and Decline buttons and a voice prompt.
- **Live capture screen (C2).** Shows the camera preview, the detected emotion label with confidence, the zone name and a small indicator that the camera is in use. A single button pauses capture, and capture stops automatically in no-capture zones.
- **Journey summary screen (C2).** A line chart of valence over time with colored bands for each zone and a list of highlights, for example the zone where the visit was most positive.
- **Recommendations screen (C2 and C4).** A short list of suggestions for the rest of the visit, each with the reason and the walking time.
- **Driver app dashboard (C3).** The "IoT Monitor" interface displays the real-time risk score, EAR, forward distance (TTC), and a live map driven by the SIM7600CE GNSS, alongside active alerts.
- **Manager dashboard (C2).** A map with a zone heatmap, filters for time range, a satisfaction trend chart, an anomaly log with evidence, recommendations and a configuration panel.
- **Operator Operator dashboard & FCM Alerts (C3 and C4).** For C4, a site view with occupancy gauges and forecasts. For C3, the fleet administrator receives mobile FCM push notifications showing a filtered list of safety alerts (type and timestamp only, preserving location privacy).

Text is concise, sentences are written in plain English with Sinhala and Tamil translations planned as future work, color is never the only carrier of meaning, and touch targets follow common mobile guidelines for minimum size.

**Key algorithms**

***Check-in emotion classification (C2))***

1. Capture a user-initiated check-in image..
2. Detect the largest face using Haar Cascade, with rotation fallback when needed..
3. Prepare normalized 100 × 100 grayscale inputs.
4. Run the custom CNN using crop scales of 0.95, 1.0 and 1.05.
5. Average the probability vectors to obtain the emotion label and confidence.
6. Associate the record with its timestamp and zone.

***Anomaly rule evaluation (C2)***

1. For each active session, maintain the smoothed valence series and the time spent in the current zone.
2. Rule A: if valence has been at or below -1 for longer than the configured duration, raise a sustained-negative anomaly with the evidence.
3. Rule B: if valence has dropped by two or more levels within 30 seconds, raise an abrupt-change anomaly with high severity.
4. For each zone, compute the rolling aggregate satisfaction; if it is below the baseline with enough records, raise a Rule C anomaly.
5. Apply de-duplication and rate limits, then dispatch to the tourist, to staff or to the recommendation engine according to type and severity.

***Adaptive repair of an itinerary (C1)***

1. Receive an event and identify the stops, times and constraints it affects.
2. Fix all completed and in-progress parts of the plan and mark the affected parts as free.
3. Run a reduced multi-objective search over the free parts, using the current conditions and the user’s weights.
4. Select a small number of non-dominated repair options that differ meaningfully, and compute their effect on cost, time and interest match.

5. Present the options by voice and screen; on acceptance, update the itinerary and publish the change; on decline, record it and apply the cool-down.

***Redistribution advice (C4)***

1. Forecast occupancy by site and hour and compare it with the comfortable threshold.
2. For each overloaded slot, enumerate candidate actions: shift earlier, shift later, alternative site or re-order.
3. Score each action by peak reduction, inconvenience and likelihood of acceptance, and select the best actions within the number of messages allowed per user.
4. Publish the advice with reasons, and record acceptance and subsequent occupancy to evaluate the effect.

## Model training configuration

**Table P.3: Training configuration of the learned models (to be confirmed by each component owner)**

| Model | Configuration | Notes |
|---|---|---|
| C2 custom CNN classifier | Conv2D, MaxPool2D, Dropout and Dense layers; 100 × 100 grayscale input; five-class output; trained on a five-class subset of RAF-DB | OpenCV Haar Cascade face detection, rotation fallback and crop-ensemble inference using scales of 0.95, 1.0 and 1.05. |
| C1 intent and entity model | Google Gemini accessed through ChatGoogleGenerativeAI; prompt-based extraction to a JSON schema; no fine-tuning, so no training data, epochs or learning rate | Two-phase chat intake (flight, then trip); session state checked at every turn |

| Model | Configuration | Notes |
|---|---|---|
| C1 NSGA-II (package optimizer) | Multi-objective route evaluation using distance, duration, road risk, weather risk, crowd pressure, attraction value and lodging value; Pareto ranking, crowding distance and weighted compromise scoring | Candidate routes are evaluated using non-dominated sorting, and the recommended package is selected from the first Pareto front using a normalized weighted compromise score. |
| C1 NSGA-II (route ranking) | Non-dominated sorting and crowding distance over the Google Routes alternatives; five minimized objectives (time, budget, road risk, weather risk, crowd exposure) | No population, generations, crossover or mutation; the candidates are the route alternatives |
| C3 Eye-State Model | MobileNetV2 backbone; MRL Eye dataset; 16 epochs; categorical cross-entropy loss | Reached 99.21% validation/test accuracy at best epoch 12 |
| C3 Yawn Detection Model | MobileNetV2 backbone; YawDD dataset; 20 epochs; categorical cross-entropy loss | Reached 99.34% validation accuracy and 99.39% test accuracy at best epoch 17 |
| C4 forecasting models | Heuristic crowd score from 0 to 100 (seasonality, Wikipedia pageview trend, holiday and weekend penalties, weather adjustment); forecasting models. | Crowd score above 80 triggers rescheduling to the early morning or the next day |